SOFTWARE DESIGN DOCUMENT

# NINAAllskySafetyMonitor

NINA safety-monitor plugin using Allsky camera image analysis and WunderStation ISYDNEY478 weather data

| | |
|---|---|
| Document | Software Design Document |
| Version | 1.12 |
| Date | 6 November 2026 |
| Author | Mark |
| Requirements baseline | requirements.md v5.8 — REQ-01 to REQ-63 |
| Notation | UML (use case, class, sequence), Rational Rose style |
| Status | Draft for review (slices 1–5 verified; slice 6 implemented, verification in progress; slice 7 harness and review tool implemented, verification in progress) |

**Revision history**

| Version | Date | Change |
|---|---|---|
| 1.0 | 1 Nov 2026 | First issue: use case model, class design, scenarios, traceability matrix. |
| 1.1 | 1 Nov 2026 | Updated to match the slice 1 code: MonitorHost and singleton settings store, IMonitoringCycle and MonitorPorts, settings read every cycle, views as XAML resource dictionaries, source-scanning coverage tests and test fakes, extra settings fields. REQ-06 rationale reworded. See §1.16. |
| 1.2 | 2 Nov 2026 | Slices 2–4 as built (acquisition, weather, decision, override, decision log, redaction) and requirements v5 working area (REQ-22, REQ-25, REQ-26, REQ-58). See §1.15. |
| 1.3 | 2 Nov 2026 | Slice 5a as built (mount, coordinates, working area, ROI, slew hold, mount watcher) and the slice 5b design (lens calibration from catalogue stars, working-area outline, enlarged image viewer); requirements v5.2 (REQ-20 fit limits, REQ-27 mount interval, REQ-59). See §1.14. |
| 1.4 | 2 Nov 2026 | Slice 5b as built and verified: frame decoding behind IFrameDecoder (WPF), darkest-channel star detection ranked by peak, the calibration helper classes, overlay content, viewer close handling, slew moves ignored by the mount watcher; generic Imager actor. See §1.13. |
| 1.5 | 2 Nov 2026 | Slice 6 design (cloud detection) for requirements v5.3: limiting magnitude 4.4, exposure-independent brightness test against a clear-sky map, sky-frame check (REQ-60), tuning recorder saving every frame with 30-day retention, Moon calculator, status metrics and history. See §1.12. |

| Version | Date | Change |
|---|---|---|
| 1.6 | 3 Nov 2026 | Slice 6 as built: CloudDetector returns the captured reference for the host to save, the tuning recorder runs after each decision, settings migration, image size recorded with the lens, block map and metric details. See §1.11. |
| 1.7 | 3 Nov 2026 | Requirements v5.4: the imager can choose that conditions which cannot be determined count as safe (REQ-50), shown as SAFE (ASSUMED) (REQ-45). See §1.10. |
| 1.8 | 3 Nov 2026 | Slice 7 design for requirements v5.5: the harness as a cross-platform console that replays saved frames for a grid of sky positions, captures its own clear-sky reference and writes results and a review workbook (REQ-42, REQ-61). See §1.9. |
| 1.9 | 3 Nov 2026 | Review tool for requirements v5.6 (REQ-62): a Python window to step through the frames picked for review and record your verdicts in review.xlsx. Harness progress on one line. See §1.8. |
| 1.10 | 3 Nov 2026 | Review tool as built: the table of rows, ↑/↓ and mouse-wheel navigation, resuming where the last review stopped, Excel recalculation on opening. See §1.7. |
| 1.11 | 5 Nov 2026 | Working-circle editor in the review tool for requirements v5.7 (REQ-63). See §1.6. |
| 1.12 | 6 Nov 2026 | Cloud advisory by default and the small-count star rule, for requirements v5.8 (REQ-28, 29, 37, 40, 45, 50). See §1.5. |

# Contents

**Contents** 3

**1 Introduction** 5

1.1 Purpose . . . 5
1.2 Scope . . . 5
1.3 References . . . 5
1.4 Conventions . . . 5
1.5 Changes in version 1.12 . . . 5
1.6 Changes in version 1.11 . . . 6
1.7 Changes in version 1.10 . . . 6
1.8 Changes in version 1.9 . . . 6
1.9 Changes in version 1.8 . . . 7
1.10 Changes in version 1.7 . . . 7
1.11 Changes in version 1.6 . . . 8
1.12 Changes in version 1.5 . . . 9
1.13 Changes in version 1.4 . . . 9
1.14 Changes in version 1.3 . . . 10
1.15 Changes in version 1.2 . . . 10
1.16 Changes in version 1.1 . . . 11

**2 Architecture** 13

2.1 Principles . . . 13
2.2 Assemblies . . . 13
2.3 NINA integration . . . 13
2.4 Threading and fail-safe behaviour . . . 13
2.5 Settings . . . 13
2.6 Traceability process . . . 14
2.7 Candidate libraries . . . 14

**3 Use case model** 15

Figure 3-1 UC-A Operation — monitoring and reporting to NINA . . . 16
Figure 3-2 UC-B Set-up, review and tuning . . . 17
Figure 3-3 UC-Ca Requirements §1–§8 → use cases . . . 18
Figure 3-4 UC-Cb Requirements §9–§13 → use cases . . . 19

**4 Class design** 20

Figure 4-1 CL-01 Assemblies and packages . . . 24
Figure 4-2 CL-02 Domain . . . 25
Figure 4-3 CL-03a Ports . . . 26
Figure 4-4 CL-03b Settings . . . 27
Figure 4-5 CL-04 Image acquisition . . . 28

Figure 4-6 CL-05 Weather . . . 29
Figure 4-7 CL-06 Geometry . . . 30
Figure 4-8 CL-07 Sky-frame check and cloud detection . . . 31
Figure 4-9 CL-08 Decision and monitoring . . . 32
Figure 4-10 CL-09 Calibration, tuning, logging and traceability . . . 33
Figure 4-11 CL-10a Plugin: NINA integration . . . 34
Figure 4-12 CL-10b Plugin: view models and views . . . 35
Figure 4-13 CL-11a Harness: offline replay . . . 36
Figure 4-14 CL-11b Tests . . . 37
Figure 4-15 CL-12 Review tool . . . 38

**5 Scenarios 39**

Figure 5-1 SD-01 Connect safety monitor . . . 40
Figure 5-2 SD-02 Monitoring cycle, normal path . . . 41
Figure 5-3 SD-03 Check the frame and detect cloud in the ROI . . . 42
Figure 5-4 SD-04 Conditions fail and clear . . . 43
Figure 5-5 SD-05 Slewing, fallback and the working area . . . 44
Figure 5-6 SD-06 Acquire image over SFTP . . . 45
Figure 5-7 SD-07 Monitor failure and idle window . . . 46
Figure 5-8 SD-08 Calibrate lens and define the working area . . . 47
Figure 5-9 SD-09 Replay saved frames offline . . . 48
Figure 5-10 SD-10 Manual override and disconnect . . . 49
Figure 5-11 SD-11 View status and edit settings . . . 50
Figure 5-12 SD-12 View the Allsky image enlarged . . . 51
Figure 5-13 SD-13 Review the replay results . . . 52
Figure 5-14 SD-14 Adjust the working circle offline . . . 53

**6 Traceability matrix 54**

# 1 Introduction

## 1.1 Purpose

This document describes the design of NINAAllskySafetyMonitor, a NINA plugin that reports whether it is safe to image. It analyses the latest image from the roof-mounted Allsky camera for cloud in the region of sky the telescope is pointing at, combines that with rain and wind data from weather station ISYDNEY478, and reports the result to NINA as a safety monitor device.

## 1.2 Scope

The design covers the use case model, the object-oriented class structure, the main and failure scenarios, and traceability from every requirement in requirements.md v5.8 to the design elements that realise it. Implementation details below class level are left to the code.

## 1.3 References

| Ref | Document |
|---|---|
| [R1] | NINAAllskySafetyMonitor Requirements, version 5.8, 6 November 2026 (requirements.md) |
| [R2] | N.I.N.A. source code, github.com/isbeorn/nina (commit 717ea13, 23 Oct 2026): IEquipmentProvider, ISafetyMonitor, ITelescopeMediator, NINA.Astrometry |
| [R3] | N.I.N.A. plugin template, github.com/isbeorn/nina.plugin.template |
| [R4] | Weather.com PWS current-observations API (response format confirmed 1 Nov 2026) |
| [R5] | Yale Bright Star Catalogue, 5th revised edition (Hoffleit and Warren, 1991; CDS V/50), public domain |

## 1.4 Conventions

- Diagrams use UML notation drawn in the Rational Rose style. Sources are PlantUML text files kept with the code.
- The dotted label at the foot of each class (for example `REQ-23 REQ-26`) lists the requirements the class implements. In code these become `[Requirement("REQ-23", "REQ-26")]` attributes.
- Defaults shown in brackets in the requirements are held in the JSON settings (§2.5); class diagrams show them as initial values.

## 1.5 Changes in version 1.12

Version 1.12 follows requirements v5.8. Replays of 20 Oct, 28 Oct and 4 Nov showed that the star test catches overcast and passing cloud but, with this camera, also reports cloud in clear sky where city glow or moonlight brightens the background: about 6% of sky positions on a clear moonless night and about 20% with a low Moon, mostly in the south-west. So cloud no longer decides safety by default: it is shown as advice, the imager pauses with Force Unsafe, and rain, wind, stale weather and the operating window still decide automatically. The small-count rule removes the "inconclusive" results under overcast.

| Diagram | Change | Reason |
|---|---|---|
| CL-03b | DecisionSettings.CloudDecidesSafety [false]; CloudDetectionSettings.ClearSkyDetectionRate [0.95], SmallCountChance [0.01], MinSmallCountStars [3]. | REQ-28, REQ-29: configurable values. |
| CL-02, CL-08, SD-04 | SafetyConditions.Deciding(cloudDecidesSafety) drops the FreshImage and NoCloud checks when cloud is advisory; SafetyDecisionEngine decides on what remains (the window alone when weather checking is also off). All checks stay in the snapshot for display and the log. | REQ-28, REQ-37, REQ-50: cloud and image problems do not change safety unless the imager chooses. |
| CL-07 | StarFractionTest: with MinSmallCountStars to MinExpectedStars−1 stars expected, cloud when the binomial chance of so few found in clear sky is below SmallCountChance, clear at MinStarFraction or above, otherwise inconclusive. | REQ-29: on 4 Nov overcast went from 67% Cloud / 33% inconclusive to 99% Cloud, and clear sky from 83% to 94% Clear. |

| Diagram | Change | Reason |
|---|---|---|
| CL-10b | StatusViewModel.CloudAdviceText/Visible beside the indicator; Options gain the "Cloud decides safety" switch and the small-count values. Force Unsafe (REQ-40) is for normal use, not only testing. | REQ-40, REQ-45. |

## 1.6 Changes in version 1.11

Version 1.11 designs the working-circle editor against requirements v5.7 (new REQ-63). Seeing the working circle on real frames showed it reaching into the trees on the east side; the editor lets the imager move and resize it on saved frames, save it to the settings used for re-analysis, re-run the harness to check, and then enter the three values in the plugin.

| Diagram | Change | Reason |
|---|---|---|
| CL-12, SD-14, UC-B, UC-Cb | Review tool: S opens SettingsWindow (centre altitude, azimuth, radius, with the values the run used). Apply draws the circle and outline on the original frame (WorkingCircleDrawing, LensModel) for any row; Save writes only the circle to settings.json through SettingsFile (backup on first save, temporary file then rename). New use case UC-21. | REQ-63: adjust the working area away from the rig and check it before it goes into the monitor. |
| CL-12 | ReviewWorkbook finds the settings file, the run's working circle and each row's original frame from the Settings sheet; FrameView shows a drawn image as well as a file. | The editor works from the run being reviewed. |
| CL-11a | ReviewWorkbookWriter records the working circle (centre alt, az, radius) in the Settings sheet. | A fixed reference for the editor, whatever settings.json holds later. |
| CL-11b | A shared table of alt/az → pixel cases (tests/data/lens-projection-cases.json) is checked by both LensProjectionTests and test_working_area; the Python tests are in tests/review. | The lens maths now exists in C# and Python and must not drift apart. |
| §2.2 | Review tool row now also names the working-circle editor. | Follows from CL-12. |

## 1.7 Changes in version 1.10

Version 1.10 brings CL-12 and SD-13 up to date with the review tool as built and tried on the first night's replay. Requirements are unchanged (v5.6).

| Diagram | Change | Reason |
|---|---|---|
| CL-12, SD-13 | ReviewTable: a table of the rows in the filter under the frame (Excel row, night, time, target, both verdicts, Agrees, metrics, why), disagreements in red; picking a line shows that frame (ReviewSession.go_to, shown_rows). | Seeing the neighbouring frames and jumping straight to one is quicker than stepping. |
| CL-12, SD-13 | Moving is ↑/↓ or the mouse wheel (was ← →), from the window or the table. | Matches the table, which lists the rows top to bottom. |
| CL-12, SD-13 | ReviewApp keeps the last row and filter in .review-state.json in the run folder and ReviewSession.restore returns to them on the next start. | A long review is done over several sittings. |
| CL-12 | ReviewWorkbook marks the workbook for full recalculation on opening. | openpyxl stores no formula results, so Agrees and the Summary sheet were blank until recalculated. |

## 1.8 Changes in version 1.9

Version 1.9 adds the review tool for requirements v5.6. A full replay picks hundreds of frames for review; clicking through them one by one in Excel is slow. The review tool shows each marked-up frame large beside its metrics and records your verdict with one key, writing it into review.xlsx so the Summary sheet still does the counting.

| Diagram | Change | Reason |
|---|---|---|
| CL-12, SD-13, UC-B, UC-Cb, CL-01 | New review tool allsky_review (tools/review): Python 3 with Tkinter, Pillow and openpyxl, for macOS and Windows. ReviewWorkbook reads the Review sheet by its headings and writes only the My verdict column, saving after every verdict (refused with a message while Excel has the workbook open). ReviewSession steps through the rows with filters (not reviewed, monitor said Cloud or Clear, disagree) and keeps the tally. ReviewApp binds the keys; FrameView shows and zooms the marked-up frame; MetricsPanel shows the metrics and counts. New use case UC-20 Review replay results. | REQ-62: frames reviewed one at a time with a single action per verdict. Python fits a small desktop tool used on the Mac; Tkinter needs no extra install. |
| CL-11b | RequirementTagScanner also reads @requirement(...) in Python files, so the review tool and its pytest tests (tests/review) count towards requirement coverage. | Every requirement keeps an implementing class and a test. |
| CL-11a | ProgressLine shows replay progress on one console line overwritten in place (a line every 30 s when redirected); FrameAnalyser.IsNight counts the frames to be judged, so the time left ignores skipped daylight frames. | Requested during the first replay: the 25-frame lines scrolled, and the estimate was distorted by daylight frames. |
| §2.2, §2.7 | Review tool added to the assemblies table; Pillow and openpyxl to the libraries. | Follows from CL-12. |

## 1.9 Changes in version 1.8

Version 1.8 designs slice 7 against requirements v5.5. The harness becomes a tool of its own, independent of the plugin but built from the same Core classes, so the two weeks of frames from the Allsky archive can be re-analysed on the Mac and the cloud thresholds tuned against the imager's own judgement.

| Diagram | Change | Reason |
|---|---|---|
| CL-11a, SD-09, CL-01 | Harness redesigned (CL-11 split into CL-11a Harness and CL-11b Tests): .NET 8 console for macOS and Windows with no NINA reference. FrameCatalog lists night folders of saved frames; FrameAnalyser runs SkyFrameCheck, the operating window and CloudDetector for every target of TargetGrid (Dec −45° to +45° every 10°, RA every 1 h) whose ROI is at least half inside the working area; results.csv gets every row. | REQ-42 v5.5: a spread of sky positions, followed through each night, without the rig or NINA. |
| CL-11a, SD-09 | ReviewSelector keeps a marked-up copy (AnnotatedFrameWriter) where a target's verdict changes or a metric is within 10% of its limit; ReviewWorkbookWriter writes review.xlsx with a My verdict drop-down, an Agrees formula and a Summary sheet. | REQ-61: results checked by eye, agreement counted in Excel. |
| CL-11a | The harness captures its own clear-sky reference from the first clear moonless frame (thresholds treated as tuned until then), or uses the saved one on request. Weather is not replayed. | Independent of the plugin's settings; no weather history exists. |
| CL-06 | SunCalculator and CoreAstroCalculator: Sun, Moon and coordinates from Core alone. | The harness has no NINA; testable against astropy. |
| CL-07 | CloudDetector keeps the last frame's block map, so several ROIs on one frame measure the background once. | Many targets per frame in the harness; also mount re-checks in the plugin. |
| CL-09, CL-11a | ReplayImageSource, ReplayCommand, ReplayClock, FixedMountStateSource and TraceabilityReportCommand removed. | The harness no longer drives MonitorService; the traceability workbook is generated with the design. |
| §2.4 | SkiaSharp (JPEG decoding and drawing) and ClosedXML (review workbook) for the harness only, both MIT. | WPF decoding is Windows-only; Excel output with formulas and a drop-down. |

## 1.10 Changes in version 1.7

Version 1.7 follows requirements v5.4. Failing to unsafe suits a roof that must close for rain, but during imaging it stops a session whenever the monitor cannot tell, for example when the image is late, too few stars are expected or the weather station is not reporting. The imager can now choose that such conditions count as safe; anything detected still stops imaging.

| Diagram | Change | Reason |
|---|---|---|
| CL-02 | ConditionCheck.Undetermined (and ConditionCheck.Unknown): a failure because the condition could not be determined. SafetyConditions.OnlyUndeterminedFailures. SafetyStatus.Assumed. | REQ-50 v5.4 separates "cannot tell" from "detected". |
| CL-02, CL-05, CL-07, CL-08 | Undetermined: no image or stale image, not a sky frame, image not decodable, lens not calibrated, cloud detection unavailable, cloud unknown (too few stars and no clear-sky reference), no weather data or data out of date (WeatherEvaluator). Detected cloud, rain and wind are never undetermined. | The imager decided that missing weather data is covered by the switch too (3 Nov 2026). |
| CL-03b, CL-08, SD-04 | DecisionSettings.UndeterminedIsSafe [false]. SafetyDecisionEngine treats only-undetermined failures as clear when it is set: the safe-after countdown runs and the result is Safe with Assumed, reason "Assumed safe: …". A failed cycle (error or timeout) has no checks, so it stays unsafe. | REQ-50; a failed cycle has checked nothing, not even that it is night. |
| CL-10b, CL-09 | Status panel: SAFE (ASSUMED) on an amber banner (IsAssumed); Options: "Safe when the monitor can't tell". Decision log: Assumed safe column. | REQ-45 v5.4: an assumed safe state is shown distinctly; REQ-44. |
| CL-02, CL-10b | MonitorSnapshot.NotASkyFrame: the panel and the viewer show a frame that is not a sky frame without the overlay. | The overlay was drawn over Allsky's notice image with the sky lens model (slice 6 check 15). |

## 1.11 Changes in version 1.6

Version 1.6 brings the design into line with the slice 6 code (375 unit tests pass; NINA verification to follow). The behaviour designed in 1.5 is unchanged; these are the differences found while building it.

| Diagram | Change | Reason |
|---|---|---|
| CL-07, SD-03 | CloudDetector.Assess returns CloudDetection: the assessment plus any clear-sky reference captured from the frame. The cycle uses it at once and hands it to MonitorPorts.ReferenceCaptured, which the plugin sets to PluginSettingsStore.SaveReference. | Core does not write the settings file; the host saves, as for calibration (REQ-57). |
| CL-07 | Reference capture runs the star test over the whole working area only when no reference exists, the thresholds are tuned, the Moon is down and the ROI is not cloudy. ClearSkyReferenceBuilder.CanCapture added. | The whole-area search takes about 2.5 s on a 2500 px frame; a normal cycle about 0.4 s. |
| CL-07 | RoiMaskBuilder keeps the working-area mask until the lens, area or image size changes; ExpectedStarPredictor takes the ROI mask; StarMatcher returns one entry per expected star; StarDetector searches only the ROI bounds. | Speed, and one place that decides which pixels are in the ROI. |
| CL-07 | BackgroundStatistics uses percentiles (non-uniformity = (P84 − P16) ÷ 2 ÷ median) instead of removing stars first. BlockStatistics stores the frame size and NaN for unmeasured blocks, written to JSON as one line of numbers per array (CompactFloatArrayConverter). | Robust without a star list; the map adds a few lines to settings.json, not thousands. |
| CL-02, CL-06 | MoonState record (position, illumination); MoonCalculator.State; CloudAssessment gains Summary and Moon; PixelRect and PixelMask.Bounds. LensCalibration records ImageWidth and ImageHeight. | The reason text and the Moon line come with the assessment; the sky-frame size check needs the calibrated size (0 for older or hand-entered lenses, check skipped). |
| CL-08, CL-09, SD-02 | TuningRecorder subscribes to SnapshotAvailable, like DecisionLogger, instead of being called inside the cycle; folders are per night (local date 12 h earlier). The cycle decodes once per image and runs the analysis off the caller's thread. StatusHistory removed: the chart view model keeps tonight's points. | Rows carry the decided state; a night is not split at midnight; one less class. |
| CL-03b | SettingsMigrator: SchemaVersion 2 moves a saved default limiting magnitude 4.0 to 4.4. | Saved files keep old defaults otherwise. |

| Diagram | Change | Reason |
|---|---|---|
| CL-09 | DecisionLogger gains stars expected and found, star fraction, relative brightness, non-uniformity ratio and the Moon. | REQ-44: the metrics behind each decision. |
| CL-10a, CL-10b, SD-11 | MetricsViewModel lines (stars, brightness, Moon, reference); HistoryChartViewModel series; SettingsViewModel cloud and tuning fields, reference status and Clear button, and Save keeps the saved reference; OverlayContent.CloudStars; viewer Expected stars toggle. | REQ-45, REQ-46; a reference captured while the Options page is open is not lost. |
| CL-11 | Fakes: FakeFrameDecoder, SyntheticCloudSky (1000 px frames with cloud patches, exposure changes and a notice image). | Cloud detection tested end to end without real frames. |

## 1.12 Changes in version 1.5

Version 1.5 designs slice 6 (cloud detection) against requirements v5.3. A test on the clear 20 October 2026 frame with the calibrated lens found every expected star in 15° ROIs in open sky, but at magnitude 4.0 many ROIs expected fewer than 8 stars, and ROIs reaching behind the trees found only about 40%. Hence the 4.4 limit and the rule that only stars inside the working area and its outline are expected.

| Diagram | Change | Reason |
|---|---|---|
| CL-07, SD-03 | Cloud detection designed in full: RoiMaskBuilder, ExpectedStarPredictor (ROI ∩ working area ∩ outline, away from the Moon), StarMatcher (tolerance from the calibration RMS), StarFractionTest, BackgroundStatistics on 32 px blocks, BrightnessTest, ClearSkyReferenceBuilder, CloudDetector. | REQ-28 to REQ-31 v5.3. |
| CL-07, CL-03b | Brightness test made exposure-independent: per-block ratio to a clear-sky map, ROI relative to the rest of the working area; non-uniformity ratio is scale-free. ClearSkyReference becomes a map (BlockStatistics). LimitingMagnitude 4.4. | Allsky auto-exposure keeps overall brightness steady; the western sky is far brighter than the east (REQ-30 v5.3). |
| CL-07, SD-03, UC-Ca | SkyFrameCheck: wrong size, or a flat background inside the working area, is "Not a sky frame" (a failed NoCloud check). | REQ-60: Allsky shows a drawn notice image when its software stops. |
| CL-06 | MoonCalculator (position and illumination) in Core; NovasAstroCalculator delegates to it. | REQ-31; testable without NINA, like CoordinateConverter. |
| CL-08, SD-02, SD-01 | MonitoringCycle decodes the frame through IFrameDecoder (MonitorPorts.FrameDecoder = WpfFrameDecoder) and runs the sky-frame check, cloud detection and tuning recorder. | The placeholder "Cloud detection not implemented yet" goes. |
| CL-09, CL-03b | TuningRecorder saves every frame and a CSV row per cycle in daily folders; deletes folders older than RetentionDays [30]. | REQ-41 v5.3. |
| CL-02, CL-10b | CloudAssessment carries relative brightness, the non-uniformity ratio and the expected/found star positions; status panel metrics and the night's history graph (OxyPlot); expected stars drawn green (found) or red (missing). | REQ-45. |

## 1.13 Changes in version 1.4

Version 1.4 records slice 5b as built and verified (all 18 checks passed on 2 November 2026; 51 of 59 requirements verified). On the 20 October 2026 frame the automatic calibration matched 227 stars with an RMS error of 2.25 px. The behaviour and requirement allocation are unchanged from 1.3.

| Diagram | Change | Reason |
|---|---|---|
| CL-03a, CL-07, CL-10b | GrayFrameDecoder in Core replaced by the IFrameDecoder port, implemented by WpfFrameDecoder in the plugin; JpegBitmap loads images for display. | Keeps Core free of an imaging library (ImageSharp not needed). |
| CL-02, CL-07 | StarDetector works on the darkest of R, G, B per pixel and sorts spots by peak; DetectedStar gains Peak and AreaPx. | The camera's noisy pixels are coloured and stars are white: 28 of the 30 brightest spots on the test frame are stars. |
| CL-09 | Calibration helpers as built: SkyStar and SkyStars (catalogue positions at the frame time), SpotIndex (grid search), StarMatch, LevenbergMarquardt (own solver). | Design decision 2 Nov 2026: own solver instead of MathNet.Numerics. |

| Diagram | Change | Reason |
|---|---|---|
| CL-10b, SD-08, SD-12 | OverlayContent record for the overlay; CalibrationMessage shown under the calibration buttons; the viewer opens after the click finishes and closes once, deferred, ignoring focus loss while closing; every viewer handler catches its own errors. | A second Close during closing crashed NINA (slice 5b check 12); the message was out of sight at the foot of the page (check 14). |
| CL-08 | MountWatcher ignores position changes while the mount is slewing. | Only the start and end of a slew matter; the log had a line every 2 s during slews. |
| UC-01, UC-02 | The Imager actor is generic. | The requirements no longer name a person. |

## 1.14 Changes in version 1.3

Version 1.3 brings the design into line with slice 5a as built and verified, and designs slice 5b. A prototype fit on a clear frame from 20 October 2026 (allsky-20261020022133.jpg) matched 33 catalogue stars with an RMS error of 2.7 px (0.15°), confirming the calibration approach; the zenith is 176 px left of the image centre and north is rotated −5.6°.

| Diagram | Change | Reason |
|---|---|---|
| CL-06 | As built: CoordinateConverter (IAU GMST, J2000 precession, about 0.01° against astropy), SkyGeometry.Destination, LensProjection.Approximate and FromHorizonRadius, MountStateInterpreter.WhyUnusable, RoiCalculator returns the reason, RoiBuilder judges the circle on the sky from 400 sample points. LensCalibration gains StarsMatched, RmsErrorDeg and Manual. | RA/Dec ↔ alt/az moved from NINA.Astrometry to Core so it is unit-tested; NovasAstroCalculator delegates to it. The ROI does not depend on the lens model. |
| CL-08, SD-01, SD-05 | MountWatcher reads the mount every 2 s and asks MonitorService to re-evaluate at once on a slew, park, tracking or position change; the cycle fetches the image at most once per image interval. SafetyDecisionEngine holds the last decision while slewing. | REQ-27 v5.2: a slew is held within seconds whatever the image interval (slice 5a check 15). |
| CL-10a | NinaMountStateSource pulls TelescopeInfo from ITelescopeMediator (imported by the provider) instead of registering as a consumer. | Simpler; the watcher and the cycle ask when they need it. |
| CL-02, CL-03b | MonitorSnapshot carries the MountState; RegionOfInterest centre is optional and has a note. MountSettings (PollIntervalS 2) and LensFitSettings (MinMatchedStars 15, MaxFitErrorDeg 0.2°, catalogue mag 4.5). | REQ-27 and REQ-20 v5.2. |
| CL-07, CL-09, SD-08 | Slice 5b: StarCatalogue embedded from the Yale Bright Star Catalogue; StarDetector removes hot pixels; Calibration package redesigned: FrameTimeReader, InitialGuessSearch, StarIdentification fallback, LensFitter, CalibrationQualityCheck, LensCalibrationSession. | REQ-19 to REQ-21 and the fit-limit warning (REQ-20). Frame time from the archive file name or the live image capture time. |
| CL-10b, SD-12 | Slice 5b: ImageViewerWindow with View mode (from the status panel; outside click or Esc closes) and Calibration mode (Save or Cancel only); cursor alt/az; overlay layers. SettingsViewModel gains calibration commands; CalibrationTemplates replaced by the viewer. | REQ-59 v5.2; the panel image is too small to click stars or draw an outline. |
| CL-09 | DecisionLogger columns for mount state, RA/Dec, reported Alt/Az, ROI alt/az, mode, % inside and note; a file started by an older version gets the current header. | REQ-06 diagnostics; slice 5a check 11. |

## 1.15 Changes in version 1.2

Version 1.2 brings the design into line with the code built and verified in slices 2 to 4, and designs the working area added in requirements v5 (to be built in slice 5).

| Diagram | Change | Reason |
|---|---|---|
| CL-02, CL-03b, CL-06, SD-05 | Horizon mask replaced by the working area: WorkingAreaDefinition (centre alt/az, radius, outline) and WorkingArea. RoiBuilder clips the 15° circle to the working area and uses the whole area when the mount position is unavailable or less than MinInsideFraction is inside. RegionOfInterest gains UsesWholeWorkingArea, InsideFraction and Note. | Requirements v5: REQ-22, REQ-25, REQ-26, REQ-58. |
| CL-03a, CL-03b | CredentialKeys; DirectSourceSettings.HostKeyAlgorithm (ecdsa-sha2-nistp256); web file name allsky.jpg; RoiSettings.MinInsideFraction. | The pinned fingerprint is the Pi's ECDSA key; image.jpg on the web server is not the current image (slice 2 checks 2 and 10). |
| CL-04, SD-06 | IImageSource gains force and Reset; FetchResult.Untrusted; IRemoteFileConnector/Session separate SSH.NET from the logic; ImageSourceFactory.For; ImageFetchTester. SSH.NET 2026.0.0. | Testable without a Pi; a host-key mismatch discards earlier images (REQ-52); advisories against 2023.0.1. |
| CL-05 | WeatherFetchResult carries the error; WeatherPoller is asked every cycle and polls when due; WeatherFetchTester; the client never puts the API key in messages. | Readable reasons; REQ-32, REQ-54. |
| CL-08, SD-02, SD-04, SD-10 | The decision engine and manual override run in MonitorService on every snapshot, including failed ones; SafetyConditionsBuilder removed (the cycle gathers the conditions); image and weather fetched in parallel; override held by MonitorHost and applied at once. | A failed cycle restarts the safe-after period; override effective immediately (REQ-38 to REQ-40). |
| CL-09 | DecisionLogger writes a daily CSV (Excel) and deletes old files; RedactingLogSink replaces RollingFileLogSink; SecretRedactor reads the current secrets each time. | REQ-44, REQ-54; NINA keeps its own text log. |
| CL-10a, CL-10b, SD-11 | WindowsCredentialStore over advapi32; PasswordBoxAssistant; Test image and Test weather; status panel image, weather line, countdown and override banner. | REQ-45, REQ-46, REQ-53. |
| CL-11 | Fakes for image sources, SFTP, HTTP, weather and credentials. | Core testable without network. |

## 1.16 Changes in version 1.1

Version 1.1 brings the design into line with the code written for slice 1 (skeleton) and with decisions taken while writing it. The behaviour and requirement allocation are unchanged.

| Diagram | Change | Reason |
|---|---|---|
| CL-02 | HorizontalCoordinate and PixelPoint are record structs. | Small values created in large numbers during star matching; structs avoid heap allocation. |
| CL-03b | Added SchemaVersion, DirectSourceSettings.Port, WeatherSettings.TimeoutS; SettingsStore gains Path, ToJson, FromJson, Clone; write-then-replace save. | Settings migration, non-standard SSH port, weather timeout (REQ-51), safe editing of a copy and crash-safe saving. |
| CL-06 | SiteLocationResolver takes the fallback as a function; a profile at 0°, 0° counts as unset. OperatingWindow takes the limit per call. | Settings are read live each cycle; NINA profiles default to 0°, 0°. |
| CL-08 | Added IMonitoringCycle and MonitorPorts; MonitorService takes interval and FailSafeGuard takes timeout as functions; MonitoringCycle reads settings each cycle. | Testable with a fake cycle; one object carries the host’s adapters; saved settings apply without reconnecting. |
| CL-10a | Added MonitorHost (single instance). PluginSettingsStore is a single instance holding the current settings. The device delegates to MonitorHost. | NINA creates the plugin, device and dock panel separately; they must share one monitor and one settings copy. |
| CL-10b | Views are XAML resource dictionaries (StatusTemplates, OptionsTemplates, CalibrationTemplates) matched by key. SettingsViewModel edits a copy, with Save and Revert. Added DelegateCommand. Duplicate Views package removed. | How NINA locates plugin views; edits apply only on Save. |

| Diagram | Change | Reason |
|---|---|---|
| CL-11 | Coverage tests scan source files; added RequirementTagScanner, RepositoryPaths, RequirementInfo, CoverageReport and the Fakes package. | The plugin assemblies cannot be loaded on the Mac; fakes make Core testable without NINA. |
| SD-01, SD-07, SD-10, SD-11 | Connect, failure, disconnect and status/settings flows go through MonitorHost; settings read every cycle; Revert added. | Follows from CL-08 and CL-10a/b. |
| §2.3, CL-06 | REQ-06 rationale reworded to match requirements.md. | Research on the AM5N home-referenced Azimuth frame. |

# 2 Architecture

## 2.1 Principles

The design is object-oriented with single-role, single-responsibility classes, one class per file. All safety logic is in a class library with no NINA references, so every rule can be unit-tested and replayed offline. NINA-specific code is confined to a thin plugin layer that adapts NINA to the library's interfaces (ports).

## 2.2 Assemblies

| Assembly | Role | Depends on |
|---|---|---|
| NINAAllskySafetyMonitor.Core | All safety logic: acquisition, weather, geometry, cloud detection, decision, monitoring loop, tuning, logging, settings, traceability attribute. | — (no NINA) |
| NINAAllskySafetyMonitor.Astrometry | Implements IAstroCalculator over NINA.Astrometry (NOVAS): sun and Moon position, RA/Dec ↔ alt/az. | Core, NINA.Astrometry |
| NINAAllskySafetyMonitor.Plugin | NINA plugin: equipment provider, safety-monitor device, adapters for mount, site, credentials and logging, MVVM views. | Core, Astrometry, NINA |
| NINAAllskySafetyMonitor.Harness | Console tool for macOS and Windows: offline replay of saved frames for a grid of sky positions, with results and a review workbook (REQ-42, REQ-61). | Core, SkiaSharp, ClosedXML |
| allsky_review (tools/review) | Python 3 / Tkinter window for macOS and Windows: steps through the frames in review.xlsx and records the imager's verdicts (REQ-62); adjusts the working circle on saved frames (REQ-63). | Pillow, openpyxl |
| NINAAllskySafetyMonitor.Tests | Unit tests per class and the requirement-coverage tests; pytest tests for the review tool in tests/review. | Core |

## 2.3 NINA integration

Confirmed against the NINA source [R2]: the plugin exports an `IEquipmentProvider<ISafetyMonitor>` through MEF. NINA gathers it with `PluginEquipmentProviders<ISafetyMonitor>` and lists the returned device in Equipment → Safety Monitor alongside other drivers. The device implements `ISafetyMonitor` (`IsSafe` plus the `IDevice` members). Connect builds and starts the monitor; Disconnect stops it, so nothing runs outside NINA (REQ-01, REQ-02).

Mount state is read from `ITelescopeMediator.GetInfo` (imported by the equipment provider) every 2 s by MountWatcher and once per cycle; site location comes from `IProfileService`. The ROI is always computed from RA/Dec. The driver's reported Alt/Az is logged for diagnostics only, because the ASI Mount driver reports Azimuth in the mount's own home-referenced frame (0° at the celestial pole), not the geographic azimuth the ASCOM standard defines (REQ-06).

NINA creates the plugin, the equipment provider (and its device) and the dock panel independently. They share one `MonitorHost` and one `PluginSettingsStore` (single instances), so there is one monitor and one copy of the settings per NINA session. Connect calls `MonitorHost.StartAsync`; Disconnect and plugin teardown call `StopAsync` (CL-10a, SD-01, SD-10).

## 2.4 Threading and fail-safe behaviour

MonitorService runs the monitoring cycle on a background task, never on NINA's UI thread. Within a cycle the image fetch and the weather poll run in parallel, each with its own timeout. Every snapshot, including one from a failed cycle, then passes through SafetyDecisionEngine (unsafe at once, safe after the clear period) and ManualOverride before it is published. Each cycle runs inside FailSafeGuard, which applies a timeout and a catch-all; any exception, hang or undeterminable state yields an Unsafe status and is logged, and nothing is thrown back to NINA (REQ-49, REQ-50). Every network operation has its own timeout (REQ-51). View models marshal snapshots to the UI dispatcher.

## 2.5 Settings

All configurable values live in one JSON file (the MonitorSettings tree, CL-03b), loaded and saved by SettingsStore, including the lens calibration, working area and clear-sky reference (REQ-55 to REQ-57). The SFTP password, if used, is held in Windows Credential Manager, and secrets are redacted from all logs (REQ-53, REQ-54). The settings file is `%LOCALAPPDATA%\NINAAllskySafetyMonitor\settings.json`, outside the NINA plugin folder, so reinstalling the plugin keeps it. Saves go to a temporary file which then replaces the old one. MonitoringCycle reads the current settings at the start of every cycle, so a connected monitor applies saved changes on its next cycle without reconnecting.

## 2.6 Traceability process

Every class and every test carries `[Requirement("REQ-xx")]`. A coverage test parses requirements.md, scans the source files for tags (so the Windows-only projects are covered when tests run on the Mac), and fails the build when a requirement has no implementing class or no test, or when a tag names an unknown requirement. The two completeness checks are skipped until the last slice; a coverage report lists the gaps meanwhile. The harness regenerates traceability.xlsx for review. Section 6 gives the design-level matrix.

## 2.7 Candidate libraries

| Need | Candidate | Notes |
|---|---|---|
| SFTP with key login and host-key pinning | SSH.NET 2026.0.0 (in use) | With BouncyCastle.Cryptography and Microsoft.Extensions.Logging.Abstractions; host-key type restricted to the pinned key |
| JPEG decode and pixel access | WPF imaging (in use) | Behind IFrameDecoder in the plugin; no extra library or licence |
| Lens-model least-squares fit | Own LevenbergMarquardt (in use) | 6 parameters (centre, rotation, c1–c3), numerical Jacobian; no extra library |
| History chart | OxyPlot (in use) | Shipped with NINA; referenced, not copied |
| Harness JPEG decoding and marked-up frames | SkiaSharp (MIT) | Cross-platform; harness only |
| Review tool image display and workbook | Pillow (MIT-CMU), openpyxl (MIT) | Python, review tool only; Tkinter ships with Python |
| Harness review workbook | ClosedXML (MIT) | Formulas and drop-down in review.xlsx; harness only |
| Star catalogue | Yale Bright Star Catalogue [R5] | Stars brighter than mag 4.5 (about 900), embedded in the Core DLL as Resources/BrightStars.csv, names for the brightest |

SSH.NET confirmed in slice 2; WPF decoding, the own solver and the catalogue in slice 5b; OxyPlot in slice 6; SkiaSharp and ClosedXML to be confirmed in slice 7.

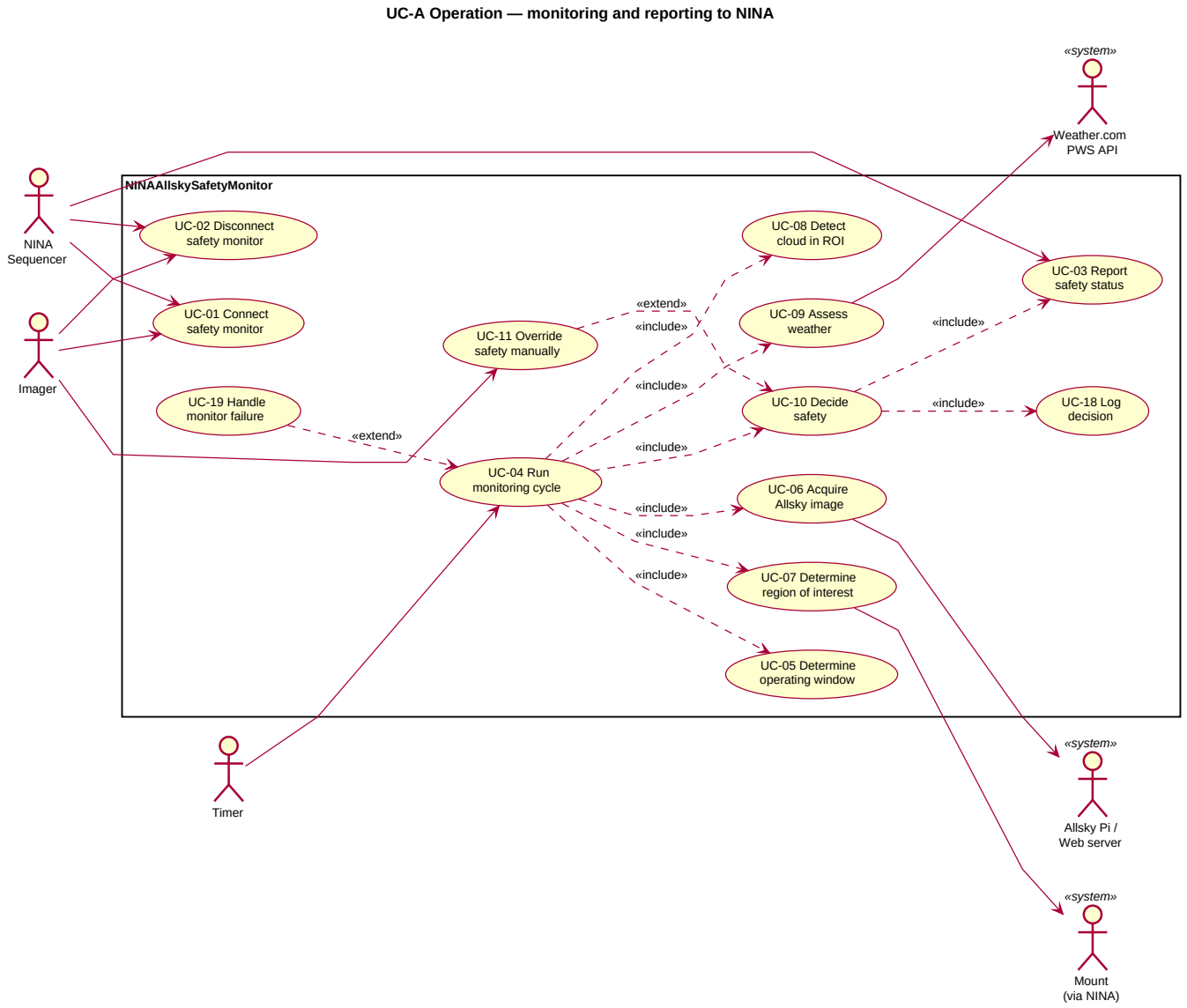
# 3 Use case model

Six actors interact with the monitor: you as the imager, the NINA sequencer, a timer that drives the monitoring cycle, and three external systems (the Allsky Pi or web server, the Weather.com PWS API, and the mount via NINA). UC-A covers operation, UC-B covers set-up and tuning, and UC-Ca/Cb trace each requirement group to a use case.

| UC | Use case | Actors | Requirements |
|---|---|---|---|
| UC-01 | Connect safety monitor | Imager, NINA | REQ-01–02, REQ-48 |
| UC-02 | Disconnect safety monitor | Imager, NINA | REQ-01–02 |
| UC-03 | Report safety status | NINA | REQ-03 |
| UC-04 | Run monitoring cycle | Timer | |
| UC-05 | Determine operating window | — | REQ-04, REQ-08–10 |
| UC-06 | Acquire Allsky image | Allsky Pi / web server | REQ-11–18, REQ-52–53, REQ-60 |
| UC-07 | Determine region of interest | Mount (via NINA) | REQ-04–07, REQ-23–27, REQ-58 |
| UC-08 | Detect cloud in ROI | — | REQ-28–31 |
| UC-09 | Assess weather | Weather.com PWS API | REQ-32–36 |
| UC-10 | Decide safety | — | REQ-37–39 |
| UC-11 | Override safety manually | Imager | REQ-40 |
| UC-12 | Calibrate lens | Imager | REQ-19–21, REQ-47 |
| UC-13 | Define working area | Imager | REQ-22, REQ-47 |
| UC-14 | View status | Imager | REQ-45, REQ-59 |
| UC-15 | Edit settings | Imager | REQ-46, REQ-53, REQ-55–57 |
| UC-16 | Record tuning data | Timer | REQ-41, REQ-43 |
| UC-17 | Replay frames offline | Imager | REQ-42, REQ-61 |
| UC-18 | Log decision | — | REQ-44, REQ-54 |
| UC-19 | Handle monitor failure | — | REQ-49–51 |
| UC-20 | Review replay results | Imager | REQ-62 |
| UC-21 | Adjust working area offline | Imager | REQ-63 |

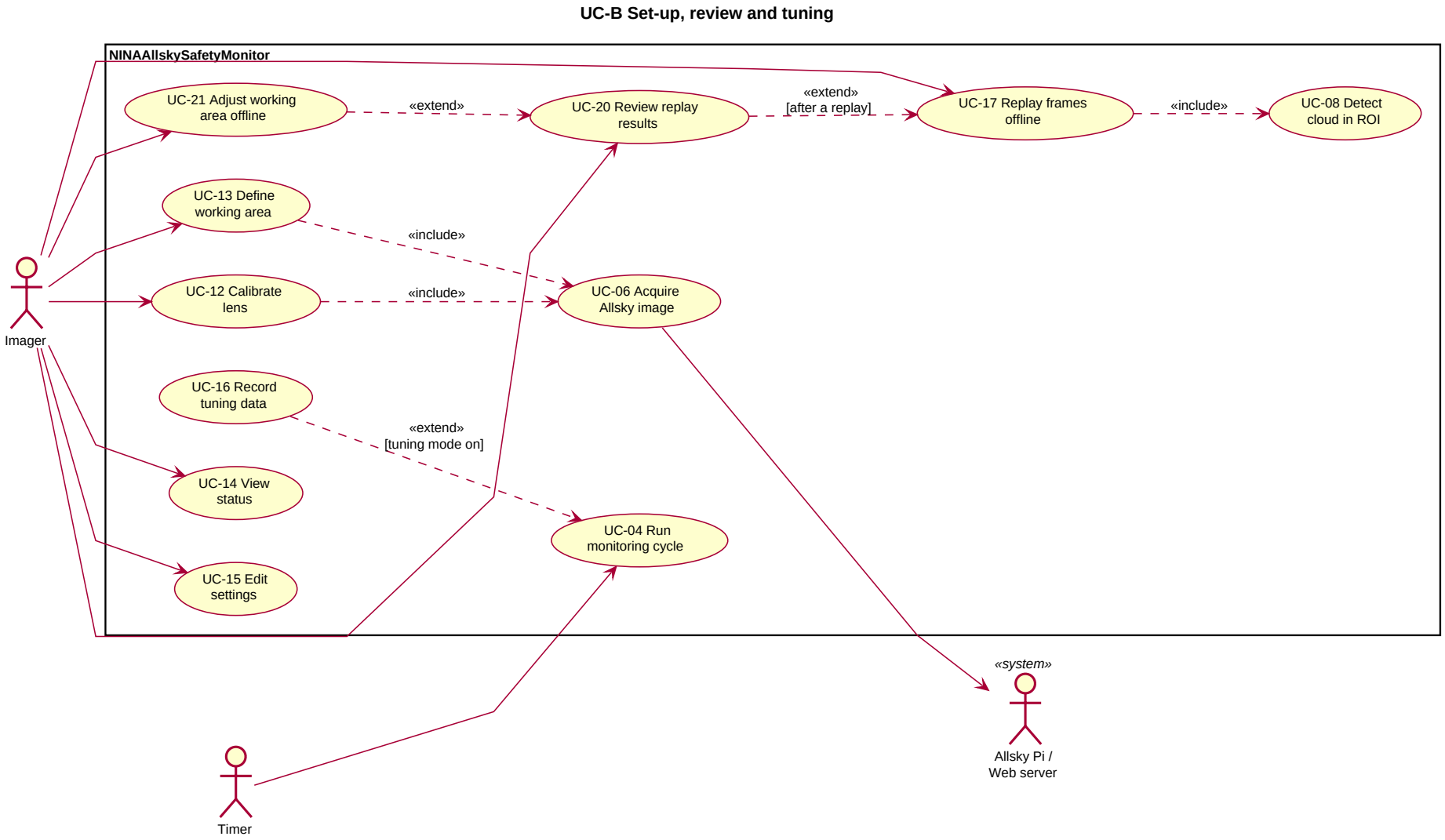
## Figure 3-1 UC-A Operation — monitoring and reporting to NINA

The timer drives UC-04, which includes the operating window, image acquisition, ROI, cloud detection, weather and decision use cases. UC-10 includes reporting to NINA and logging. Manual override and failure handling extend the flow.



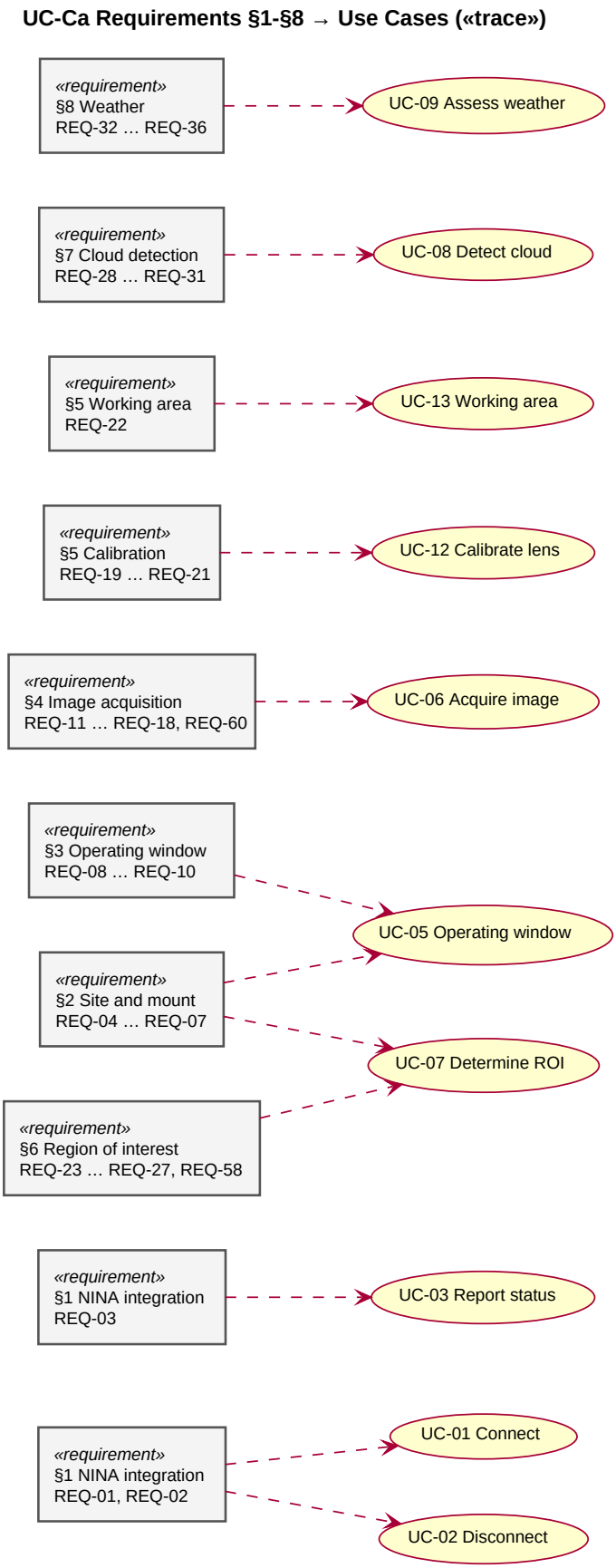
## Figure 3-2 UC-B Set-up, review and tuning

Calibrating the lens and defining the working area both use a live image. Tuning-data recording extends the monitoring cycle when tuning mode is on; offline replay reuses cloud detection, its results are then reviewed frame by frame, and the working area can be adjusted on saved frames.



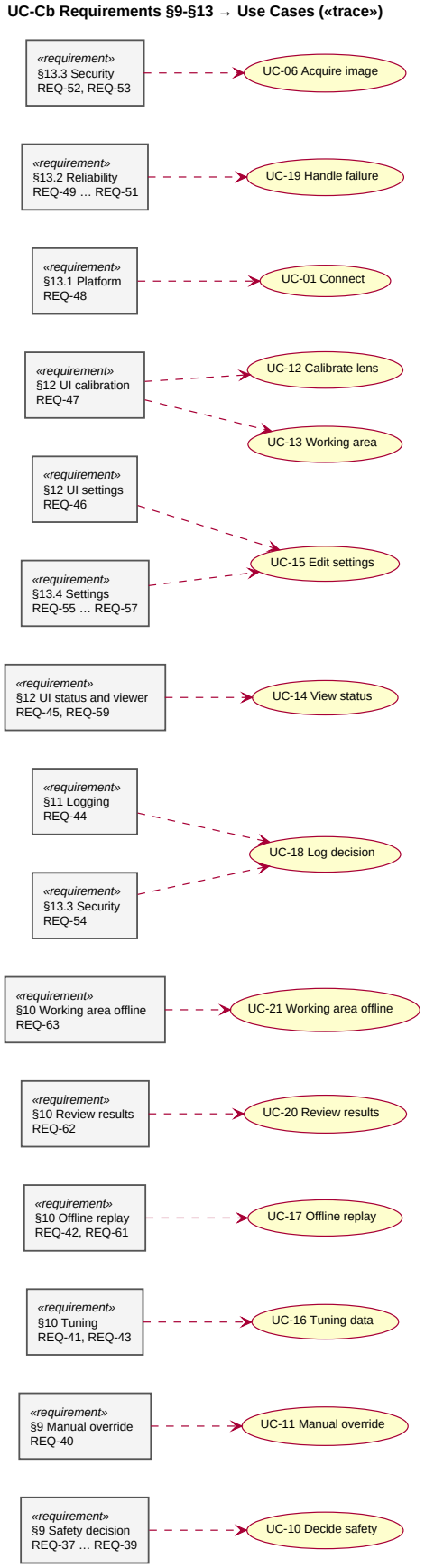
## Figure 3-3 UC-Ca Requirements §1–§8 → use cases

Functional requirement groups for NINA integration, site and mount, operating window, acquisition, calibration, ROI, cloud detection and weather.



# Figure 3-4 UC-Cb Requirements §9–§13 → use cases

Safety decision, tuning, logging, user interface and the non-functional groups.



# 4 Class design

Fifteen class diagrams describe the design package by package. Table 4-1 lists every class that implements a requirement, with its diagram.

Table 4-1 Classes and the requirements they implement

| Diagram | Class | Requirements |
|---|---|---|
| CL02 | CloudAssessment | REQ-28, REQ-45 |
| CL02 | ConditionCheck | REQ-37, REQ-50 |
| CL02 | MonitorSnapshot | REQ-41, REQ-45, REQ-60 |
| CL02 | MoonState | REQ-31 |
| CL02 | MountState | REQ-05–07 |
| CL02 | RegionOfInterest | REQ-23, REQ-25–26, REQ-58 |
| CL02 | SafetyConditions | REQ-28, REQ-37, REQ-50 |
| CL02 | SafetyStatus | REQ-03, REQ-39–40, REQ-43, REQ-45, REQ-50 |
| CL02 | SiteLocation | REQ-04 |
| CL02 | SkyImage | REQ-11, REQ-15 |
| CL02 | WeatherObservation | REQ-33 |
| CL03a | CredentialKeys | REQ-53 |
| CL03a | IAstroCalculator | REQ-08, REQ-24, REQ-31 |
| CL03a | ICredentialStore | REQ-53 |
| CL03a | IFrameDecoder | REQ-20 |
| CL03a | IMountStateSource | REQ-05 |
| CL03a | ISiteLocationSource | REQ-04 |
| CL03b | CalibrationData | REQ-57 |
| CL03b | CloudDetectionSettings | REQ-29–31, REQ-43 |
| CL03b | DecisionSettings | REQ-28, REQ-39, REQ-50 |
| CL03b | LensFitSettings | REQ-20 |
| CL03b | MonitorSettings | REQ-55 |
| CL03b | MountSettings | REQ-27 |
| CL03b | RoiSettings | REQ-23, REQ-58 |
| CL03b | SettingsMigrator | REQ-29, REQ-55 |
| CL03b | SettingsStore | REQ-55–56 |
| CL03b | SettingsValidator | REQ-22, REQ-27, REQ-46, REQ-58 |
| CL03b | TuningSettings | REQ-41 |
| CL04 | FetchResult | REQ-16, REQ-52 |
| CL04 | HostKeyMismatchException | REQ-52 |
| CL04 | HostKeyVerifier | REQ-52 |
| CL04 | IImageSource | REQ-11 |
| CL04 | ImageAcquirer | REQ-14, REQ-17–18 |
| CL04 | ImageFetchTester | REQ-46 |
| CL04 | ImageFreshnessEvaluator | REQ-15, REQ-18, REQ-50 |
| CL04 | ImageSourceFactory | REQ-11 |
| CL04 | JpegIntegrityChecker | REQ-17 |
| CL04 | SftpConnector | REQ-12–13, REQ-51–53 |
| CL04 | SftpFileSession | REQ-11 |

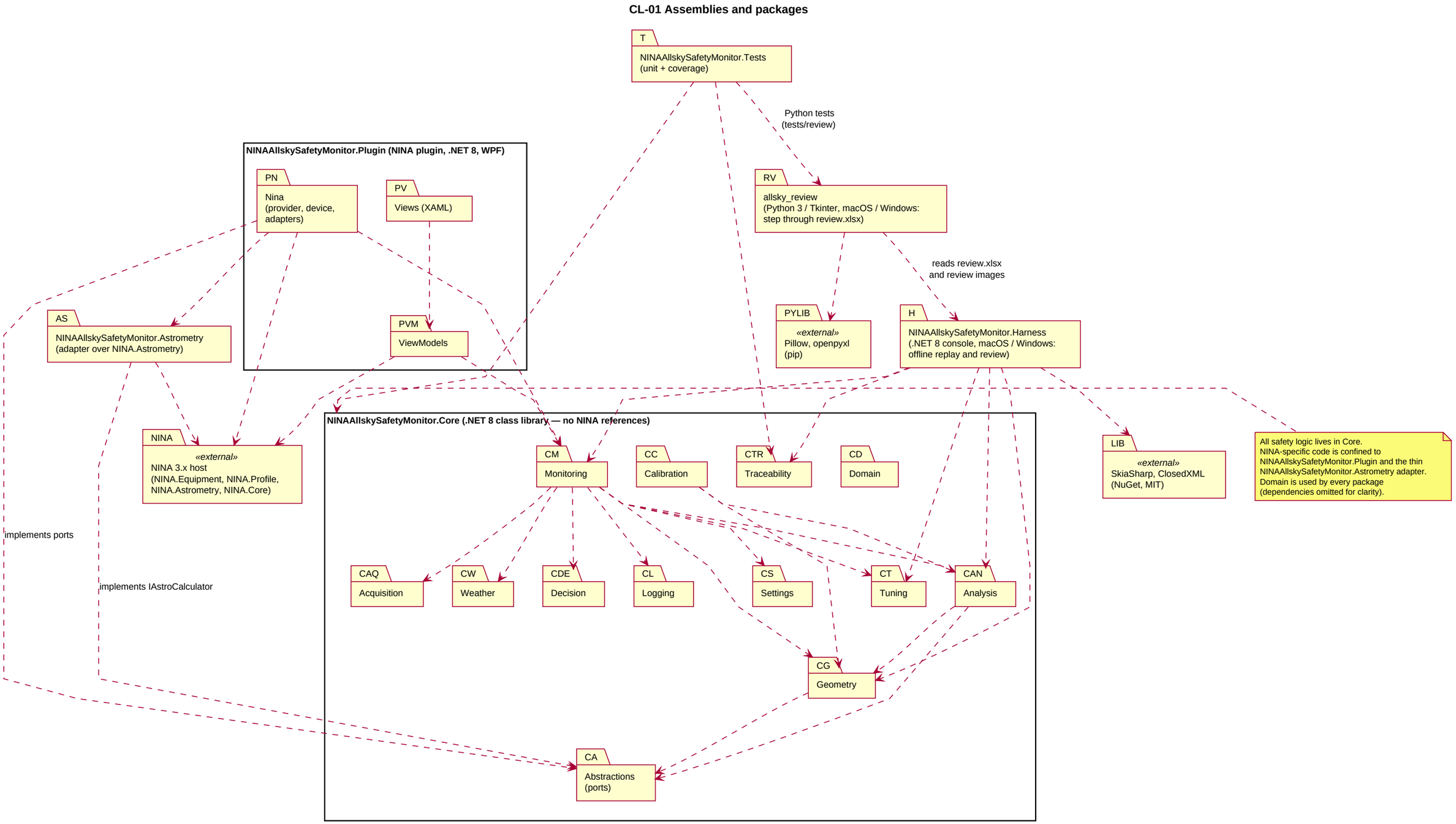
| Diagram | Class | Requirements |
|---|---|---|
| CL04 | SftpImageSource | REQ-11, REQ-15–16, REQ-51 |
| CL04 | WebImageSource | REQ-11, REQ-15, REQ-51 |
| CL05 | IWeatherProvider | REQ-32 |
| CL05 | PwsObservationParser | REQ-33 |
| CL05 | PwsWeatherClient | REQ-32, REQ-51, REQ-54 |
| CL05 | WeatherEvaluator | REQ-34–36, REQ-50 |
| CL05 | WeatherFetchTester | REQ-46 |
| CL05 | WeatherPoller | REQ-32 |
| CL06 | CoordinateConverter | REQ-24 |
| CL06 | CoreAstroCalculator | REQ-08, REQ-24, REQ-31, REQ-42 |
| CL06 | LensCalibration | REQ-19, REQ-60 |
| CL06 | LensProjection | REQ-19, REQ-24 |
| CL06 | MoonCalculator | REQ-31 |
| CL06 | MountStateInterpreter | REQ-07, REQ-26 |
| CL06 | OperatingWindow | REQ-08–10 |
| CL06 | RoiBuilder | REQ-23, REQ-25–26, REQ-58 |
| CL06 | RoiCalculator | REQ-06, REQ-23–24, REQ-26 |
| CL06 | SiteLocationResolver | REQ-04 |
| CL06 | SunCalculator | REQ-08 |
| CL06 | WorkingArea | REQ-22, REQ-25 |
| CL06 | WorkingAreaDefinition | REQ-22, REQ-57 |
| CL07 | BackgroundStatistics | REQ-30 |
| CL07 | BrightnessTest | REQ-30–31 |
| CL07 | ClearSkyReference | REQ-30, REQ-57 |
| CL07 | ClearSkyReferenceBuilder | REQ-30 |
| CL07 | CloudDetection | REQ-28, REQ-30 |
| CL07 | CloudDetector | REQ-28 |
| CL07 | ExpectedStarPredictor | REQ-29, REQ-31 |
| CL07 | MoonBrightnessAdjuster | REQ-31 |
| CL07 | RoiMaskBuilder | REQ-25, REQ-29 |
| CL07 | SkyFrameCheck | REQ-60 |
| CL07 | StarCatalogue | REQ-20, REQ-29 |
| CL07 | StarDetector | REQ-20, REQ-29 |
| CL07 | StarFractionTest | REQ-29 |
| CL07 | StarMatcher | REQ-29 |
| CL08 | FailSafeGuard | REQ-49–51 |
| CL08 | ManualOverride | REQ-40 |
| CL08 | MonitorService | REQ-02, REQ-27, REQ-38, REQ-40, REQ-49 |
| CL08 | MonitoringCycle | REQ-05, REQ-10, REQ-14, REQ-28, REQ-37, REQ-60 |
| CL08 | MountWatcher | REQ-05, REQ-27 |
| CL08 | SafetyDecisionEngine | REQ-03, REQ-27–28, REQ-37–39, REQ-50 |
| CL08 | SafetyHysteresis | REQ-38–39 |
| CL08 | SlewHoldPolicy | REQ-27 |
| CL09 | CalibrationQualityCheck | REQ-20 |

| Diagram | Class | Requirements |
|---|---|---|
| CL09 | DecisionLogger | REQ-06, REQ-44, REQ-54 |
| CL09 | FrameTimeReader | REQ-20 |
| CL09 | InitialGuessSearch | REQ-20 |
| CL09 | LensCalibrationSession | REQ-19–20 |
| CL09 | LensFitter | REQ-19–20 |
| CL09 | MetricsCsvWriter | REQ-41 |
| CL09 | RedactingLogSink | REQ-54 |
| CL09 | SecretRedactor | REQ-54 |
| CL09 | SkyStars | REQ-20, REQ-24 |
| CL09 | TuningRecorder | REQ-41 |
| CL09 | TuningStateEvaluator | REQ-43 |
| CL10a | AllskySafetyMonitorDevice | REQ-01–03, REQ-49 |
| CL10a | AllskySafetyMonitorProvider | REQ-01, REQ-05 |
| CL10a | AllskySafetyPlugin | REQ-46, REQ-48 |
| CL10a | MonitorHost | REQ-02–03, REQ-05 |
| CL10a | NinaMountStateSource | REQ-05–07 |
| CL10a | NinaSiteLocationSource | REQ-04 |
| CL10a | NovasAstroCalculator | REQ-08, REQ-24, REQ-31 |
| CL10a | PluginSettingsStore | REQ-30, REQ-55–57 |
| CL10a | WindowsCredentialStore | REQ-53 |
| CL10b | CalibrationViewModel | REQ-19–22, REQ-47, REQ-57 |
| CL10b | HistoryChartViewModel | REQ-45 |
| CL10b | ImageViewerViewModel | REQ-21, REQ-59 |
| CL10b | MetricsViewModel | REQ-45 |
| CL10b | OverlayImageComposer | REQ-21–23, REQ-45 |
| CL10b | SettingsViewModel | REQ-22–23, REQ-27, REQ-46, REQ-53, REQ-55, REQ-58 |
| CL10b | StatusViewModel | REQ-40, REQ-43, REQ-45, REQ-58–59 |
| CL10b | WpfFrameDecoder | REQ-20 |
| CL11a | AnnotatedFrameWriter | REQ-61 |
| CL11a | FrameAnalyser | REQ-42 |
| CL11a | FrameCatalog | REQ-42 |
| CL11a | Program | REQ-42 |
| CL11a | ProgressLine | REQ-42 |
| CL11a | ReplayOptions | REQ-42, REQ-61 |
| CL11a | ResultsCsvWriter | REQ-42, REQ-61 |
| CL11a | ReviewSelector | REQ-61 |
| CL11a | ReviewWorkbookWriter | REQ-61 |
| CL11a | SkiaFrameDecoder | REQ-42 |
| CL11a | TargetGrid | REQ-42 |
| CL12 | FrameView | REQ-62 |
| CL12 | LensModel | REQ-63 |
| CL12 | MetricsPanel | REQ-62 |
| CL12 | ReviewApp | REQ-62–63 |
| CL12 | ReviewRow | REQ-62 |

| Diagram | Class | Requirements |
|---|---|---|
| CL12 | ReviewSession | REQ-62 |
| CL12 | ReviewTable | REQ-62 |
| CL12 | ReviewWorkbook | REQ-61–63 |
| CL12 | RowFilter | REQ-62 |
| CL12 | SettingsFile | REQ-63 |
| CL12 | SettingsWindow | REQ-63 |
| CL12 | Tally | REQ-62 |
| CL12 | WorkingArea | REQ-63 |
| CL12 | WorkingCircleDrawing | REQ-63 |

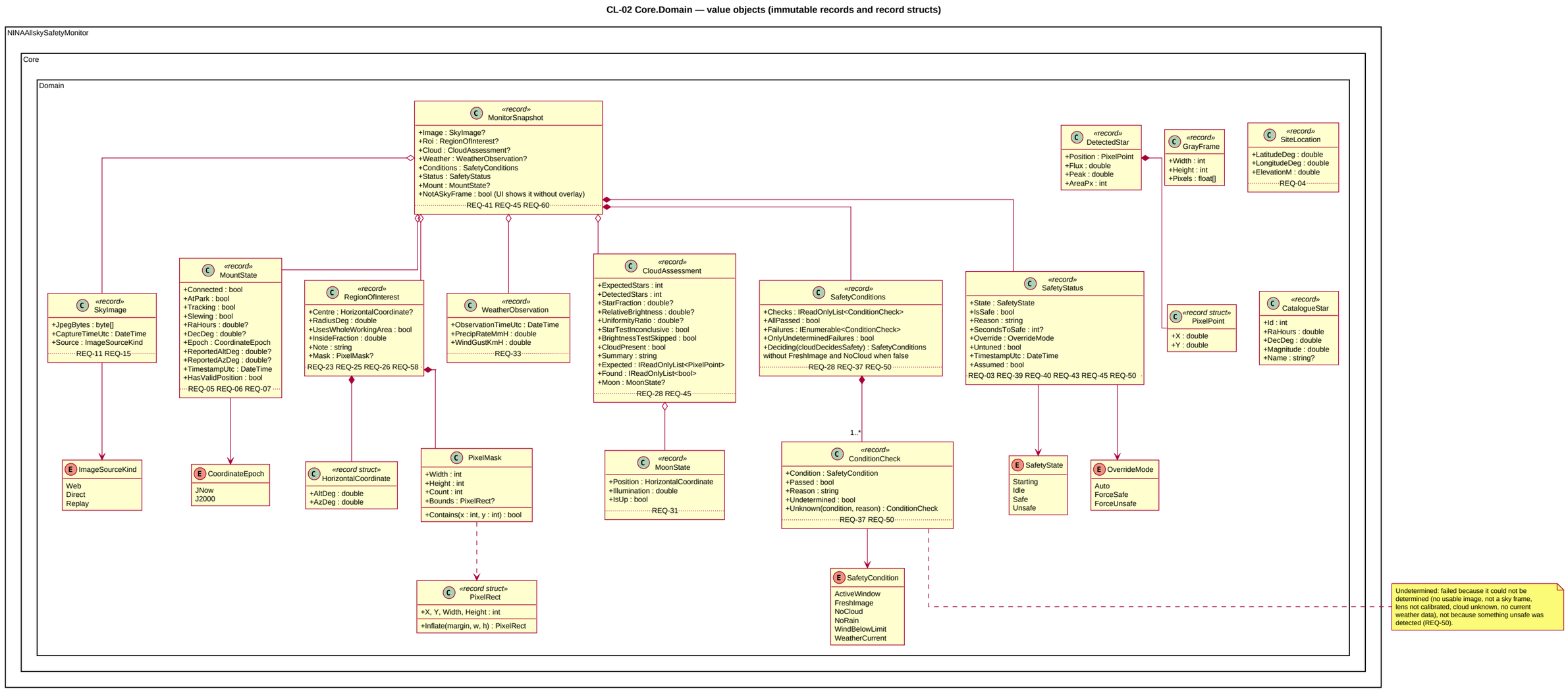
## Figure 4-1 CL-01 Assemblies and packages

Core holds all safety logic and has no NINA references. Domain types are used by every package; those dependencies are omitted for clarity.



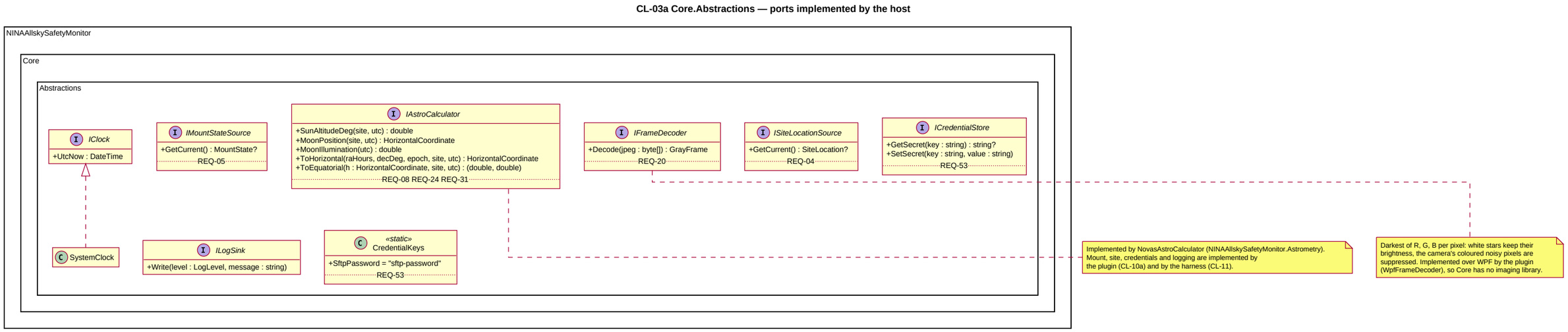
# Figure 4-2 CL-02 Domain

Immutable value objects passed between components: images, mount state, ROI, assessments, conditions, safety status and the snapshot used by the UI and tuning.



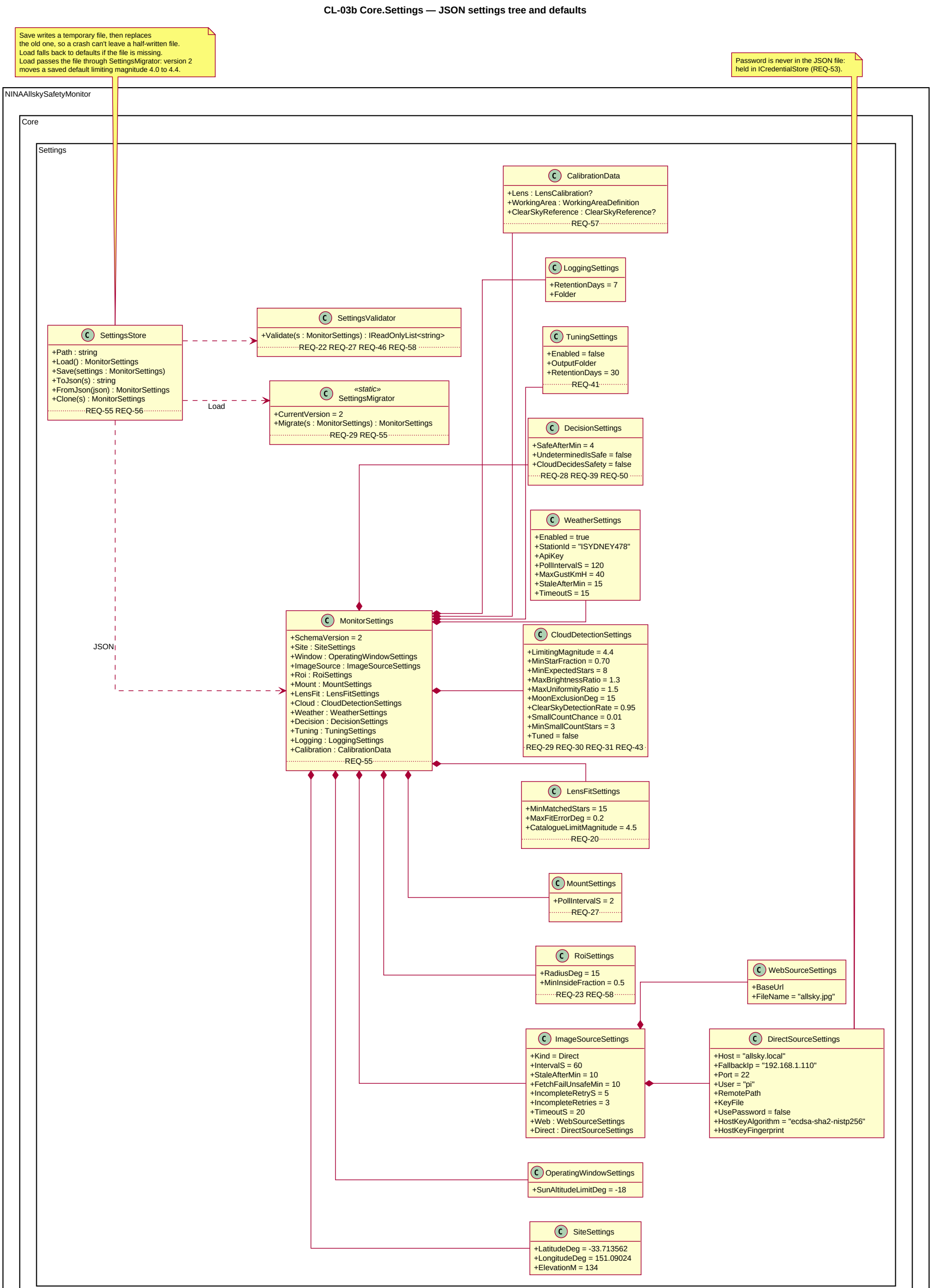
## Figure 4-3 CL-03a Ports

Interfaces Core needs from its host. The plugin implements them over NINA; the harness implements them for offline replay.



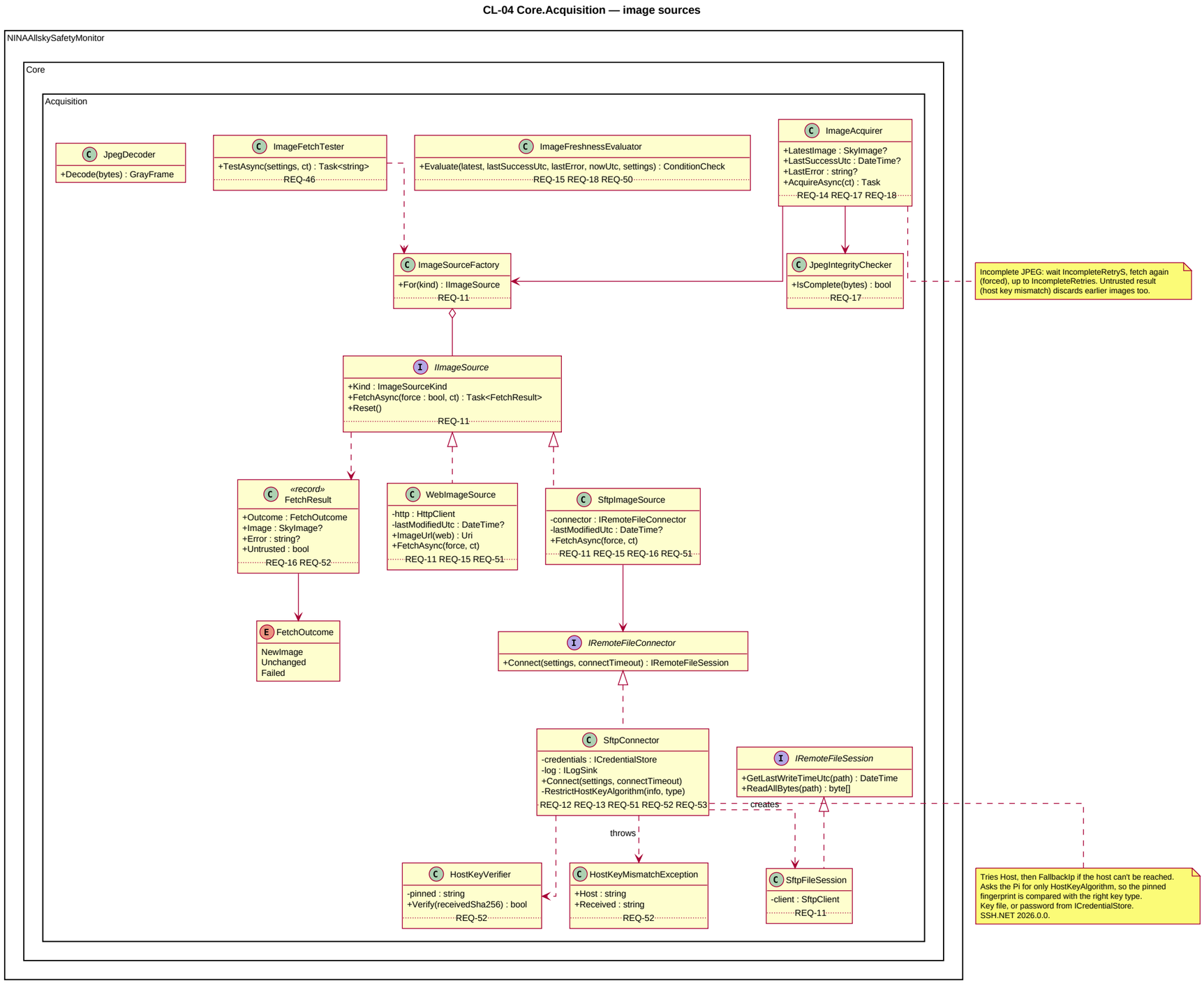
## Figure 4-4 CL-03b Settings

The JSON settings tree with its defaults. Initial values match the bracketed defaults in the requirements.



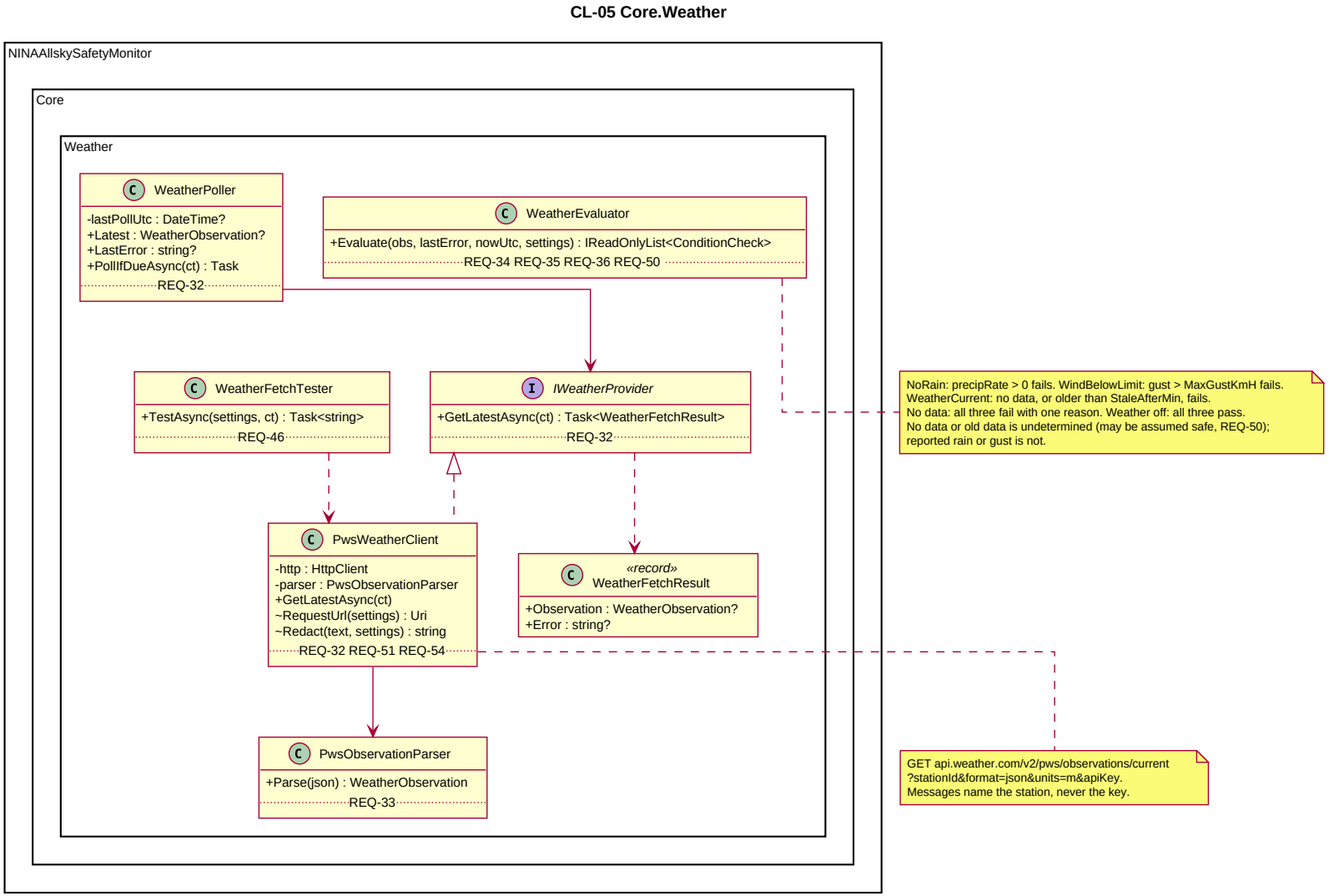
## Figure 4-5 CL-04 Image acquisition

Web and SFTP sources behind one interface, SSH.NET kept behind IRemoteFileConnector, host-key type and fingerprint pinning, the incomplete-JPEG retry and the freshness rule.



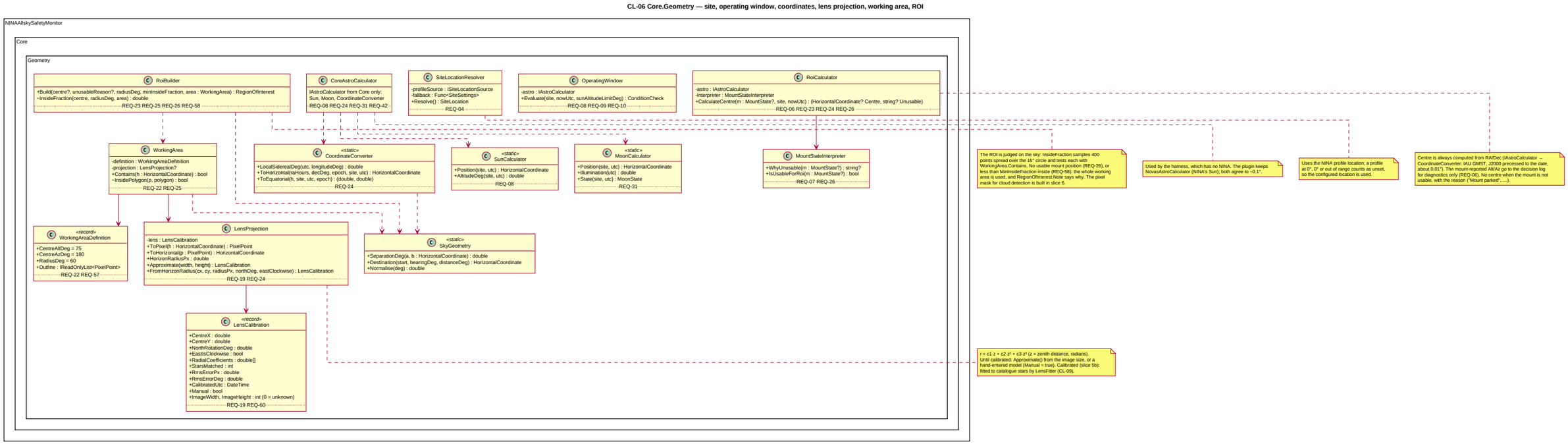
## Figure 4-6 CL-05 Weather

PWS client and parser (the API key never appears in messages), a poller asked every cycle that reads the station every 2 minutes, and the evaluator producing the rain, wind and data-age checks.



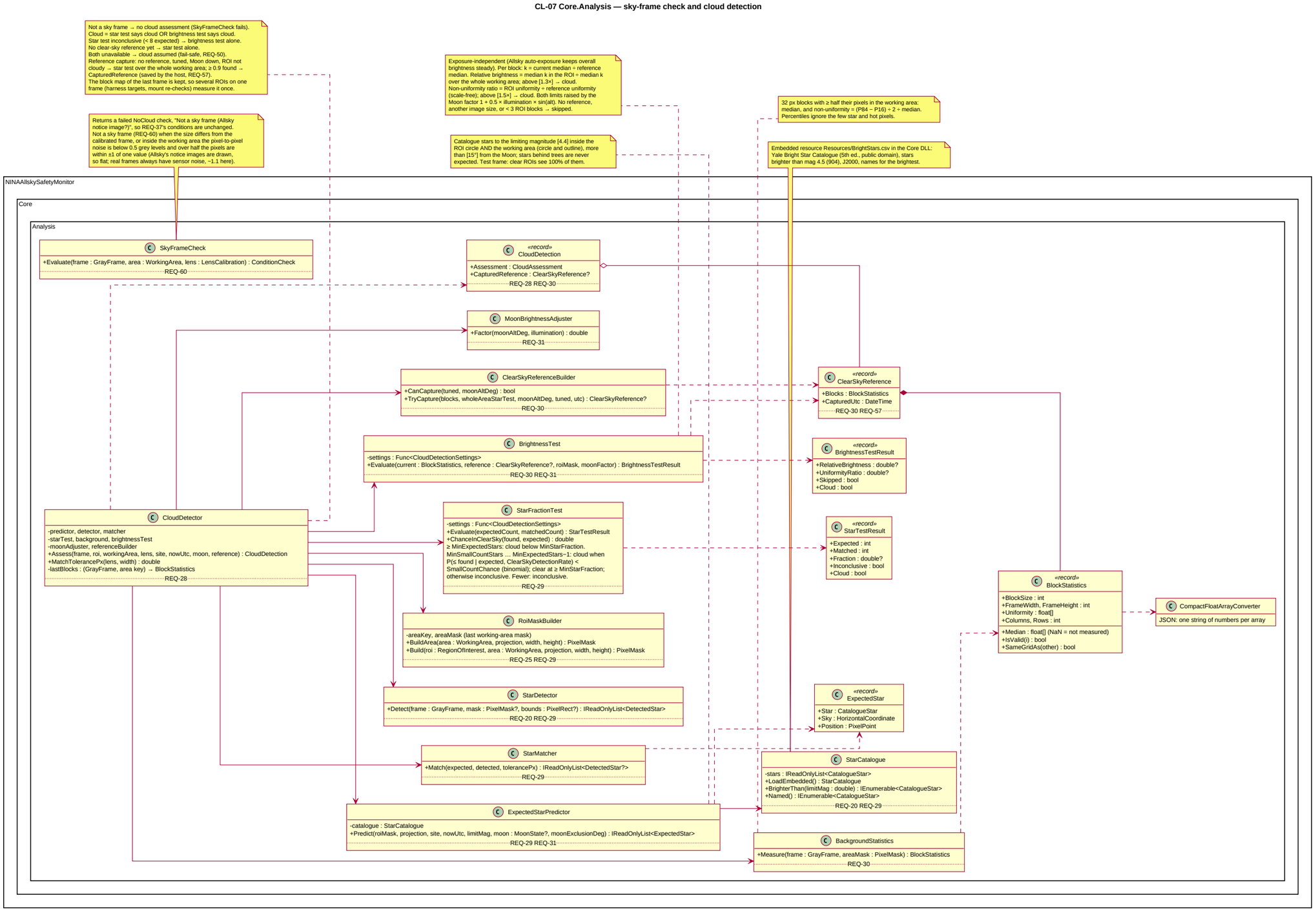
# Figure 4-7 CL-06 Geometry

Site resolution, operating window, fisheye lens projection, the working area, and the ROI built from the mount’s RA/Dec and clipped to the working area, or the whole working area as fallback.



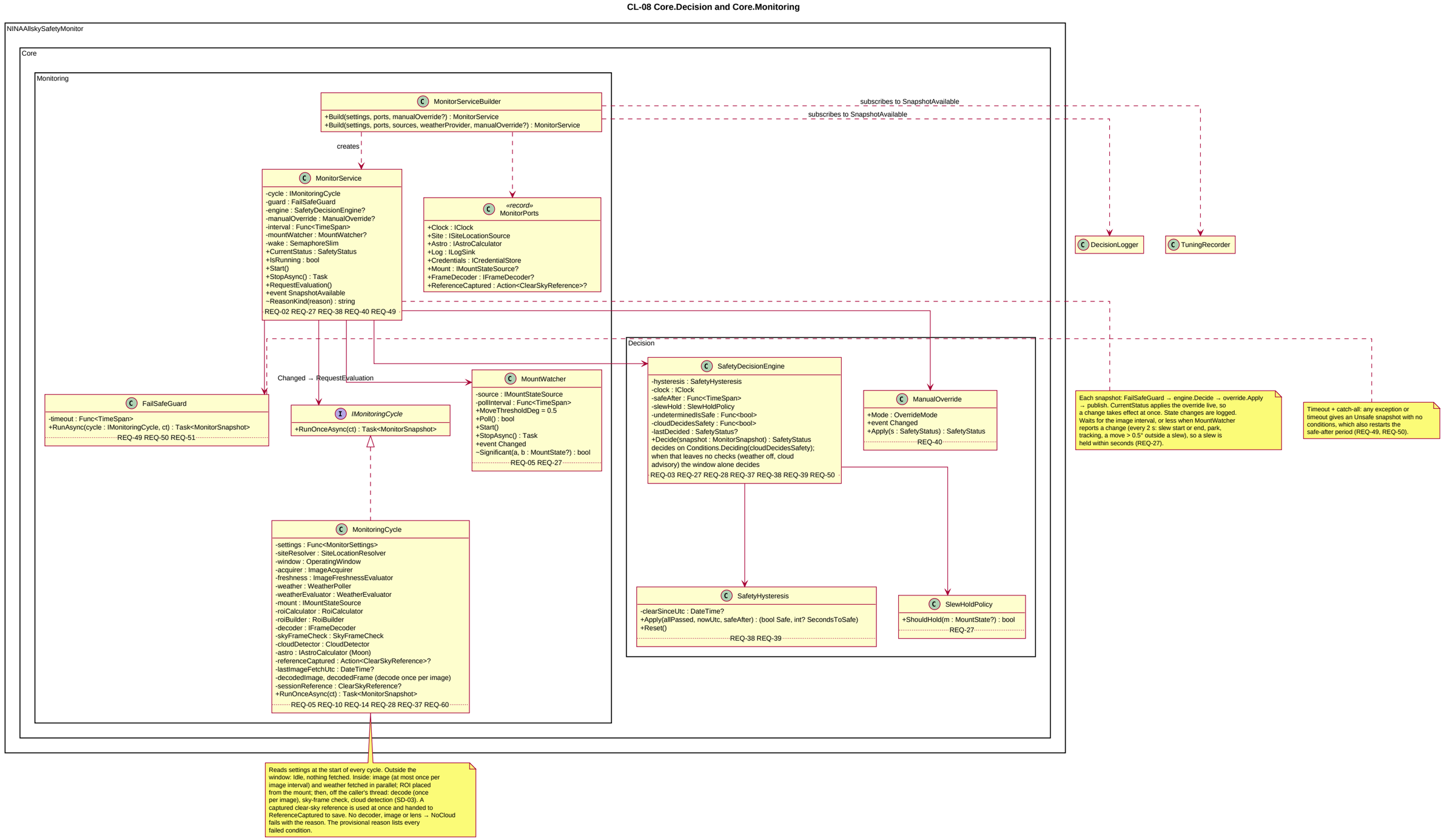
## Figure 4-8 CL-07 Sky-frame check and cloud detection

The sky-frame check; catalogue star prediction, detection and matching; the star-fraction and brightness tests against the clear-sky block map; the Moon adjustment and reference capture, combined by CloudDetector.



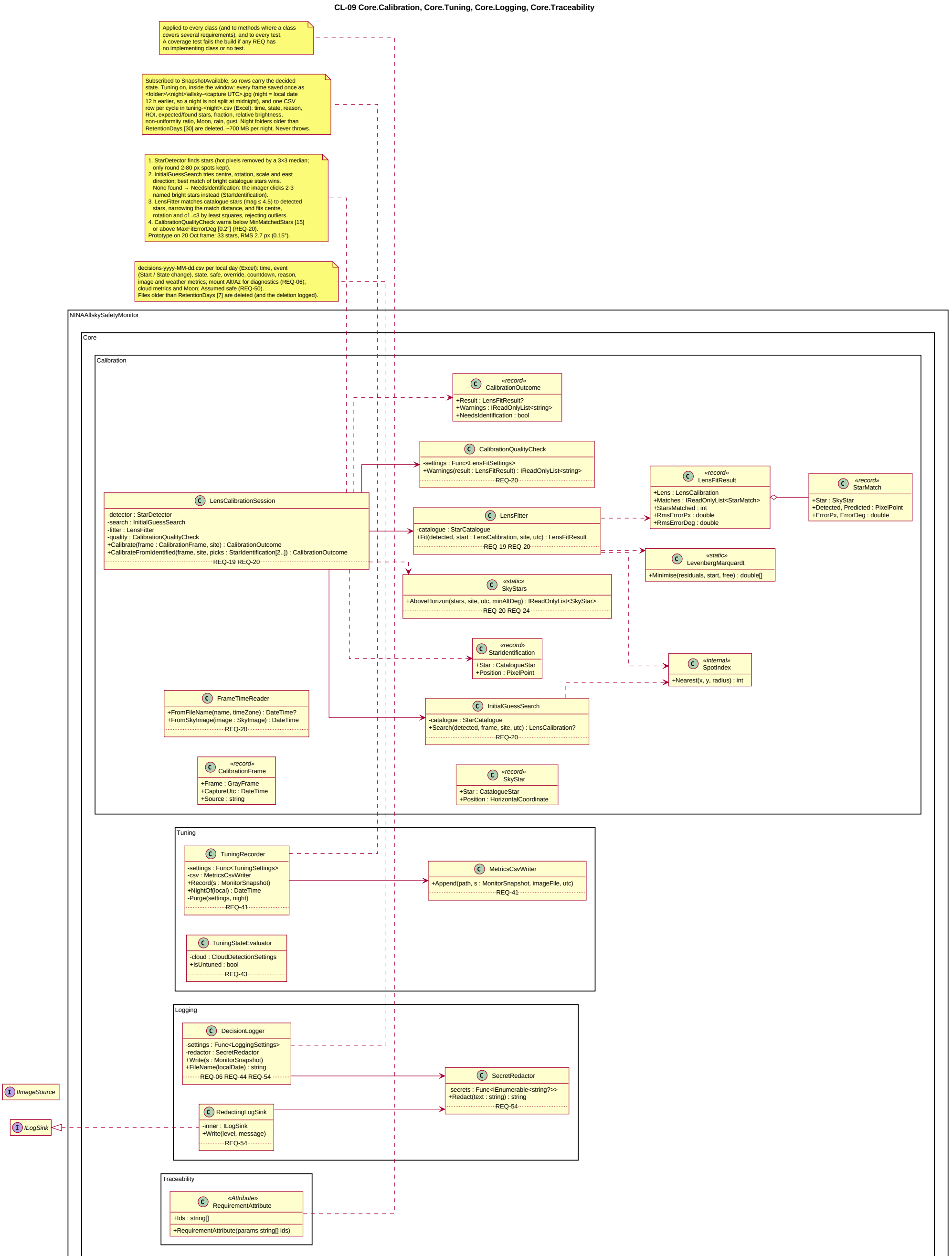
## Figure 4-9 CL-08 Decision and monitoring

MonitorService runs each cycle inside FailSafeGuard, then the decision engine (hysteresis, slew hold) and the manual override, then publishes. The builder composes everything and subscribes the decision log.



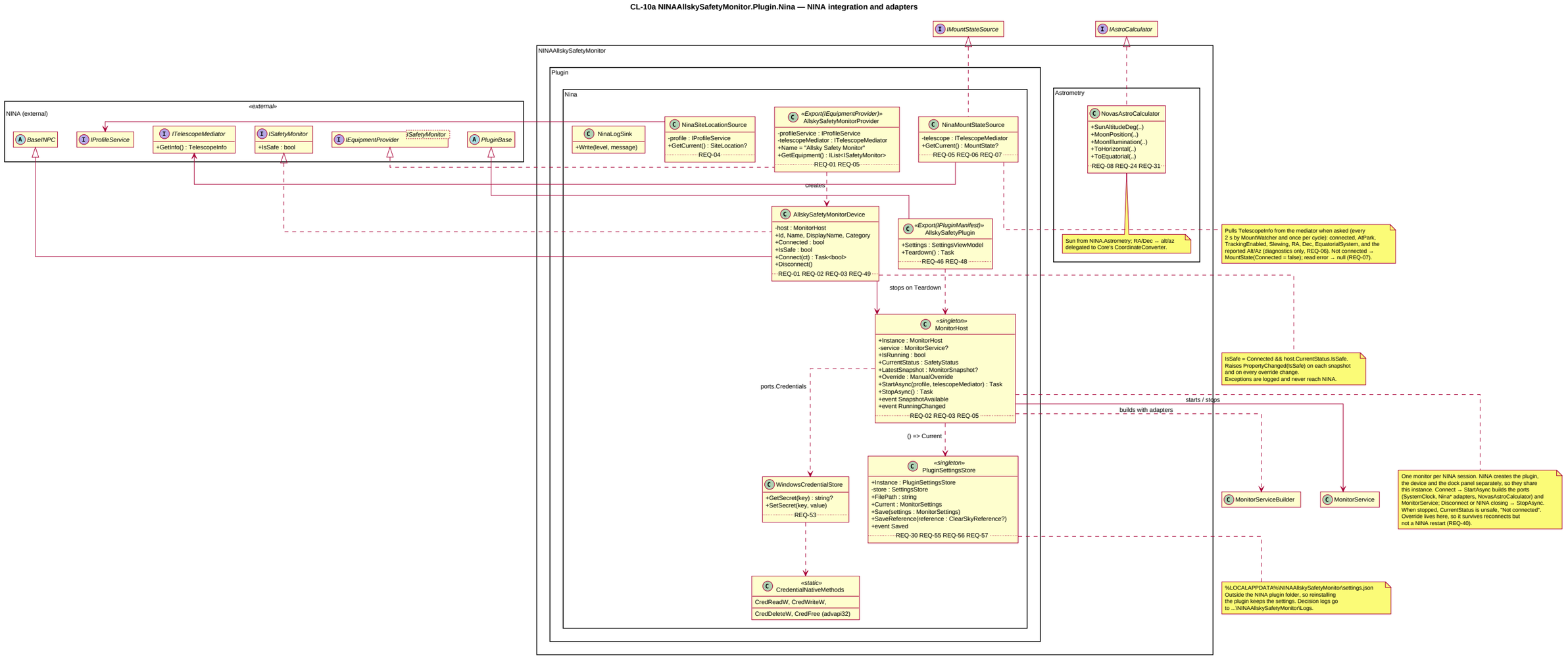
# Figure 4-10 CL-09 Calibration, tuning, logging and traceability

Lens calibration from catalogue stars (automatic start, identified stars as fallback, fit-quality warnings), the tuning recorder and replay source, the daily CSV decision log, secret redaction and the Requirement attribute.



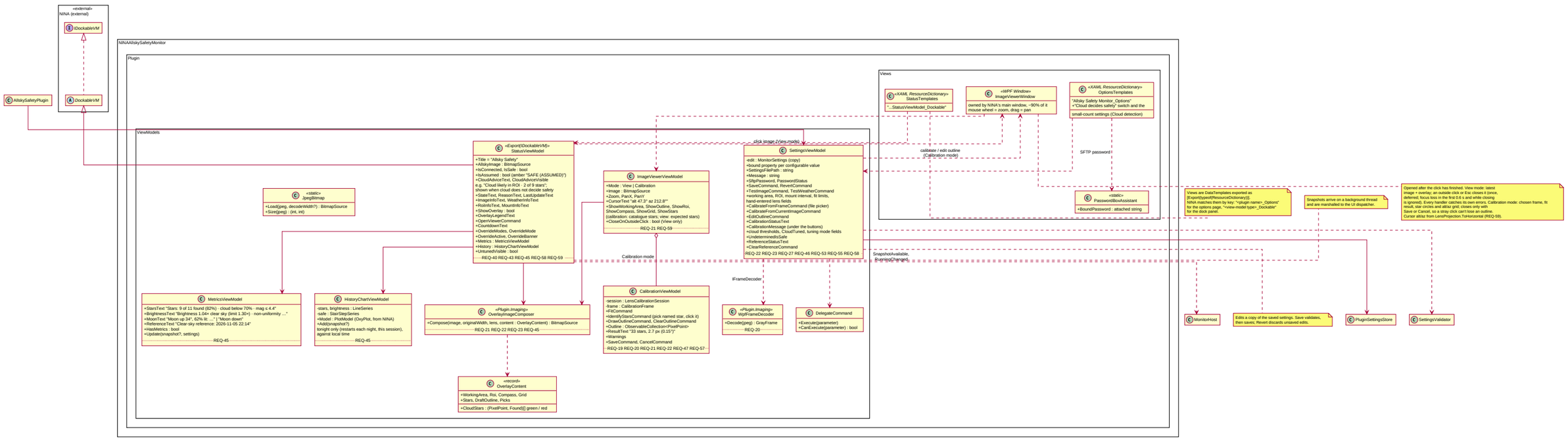
# Figure 4-11 CL-10a Plugin: NINA integration

Equipment provider, safety-monitor device, the shared MonitorHost and settings store, and the NINA adapters that implement Core’s ports.



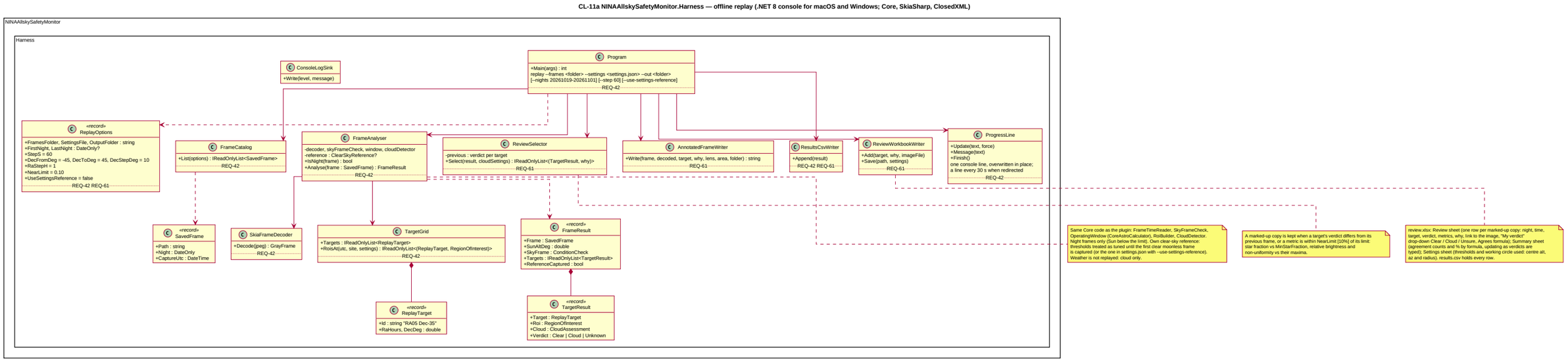
# Figure 4-12 CL-10b Plugin: view models and views

MVVM layer inside NINA: the status dock panel, settings page, and the enlarged image viewer with its view and calibration modes.



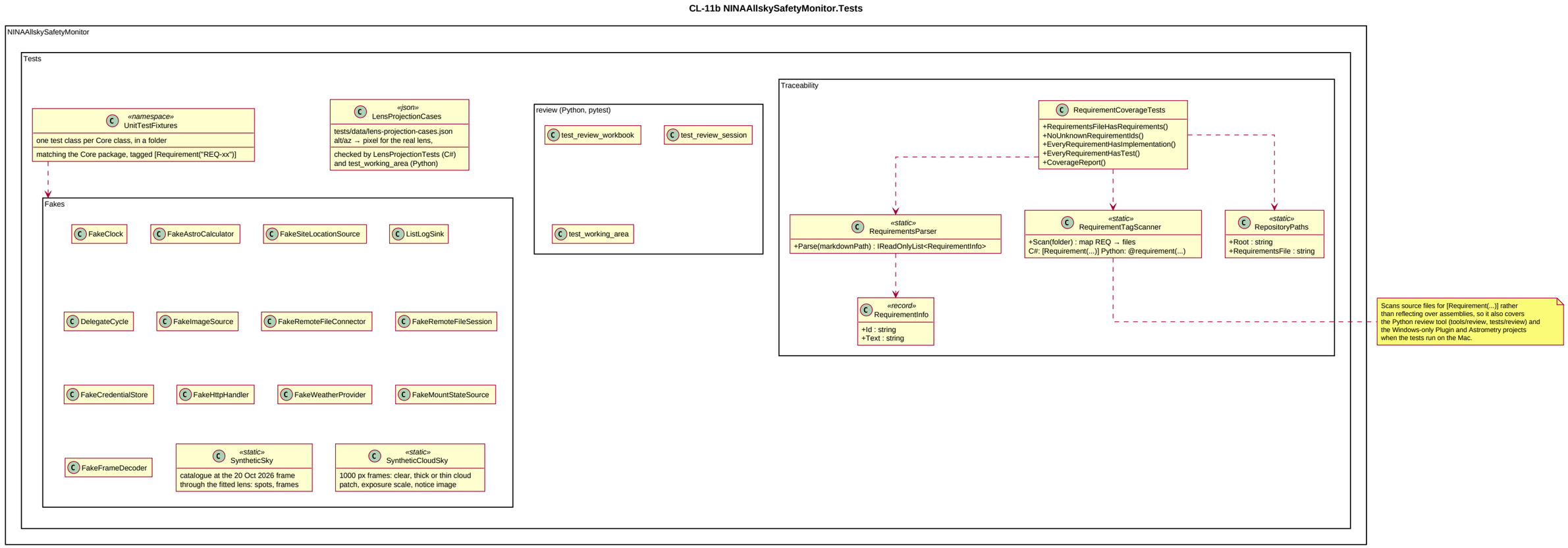
# Figure 4-13 CL-11a Harness: offline replay

The offline replay tool: frame catalogue, target grid, frame analyser, review selection, marked-up frames, results and the review workbook.



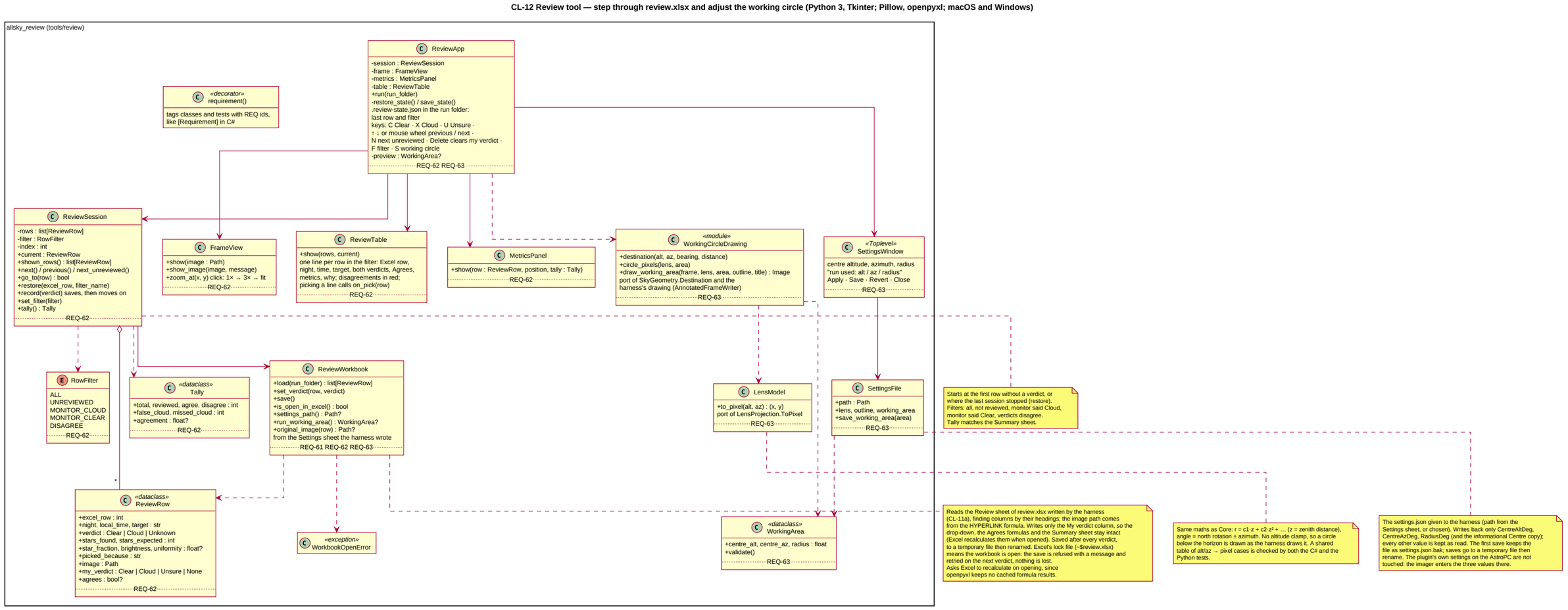
## Figure 4-14 CL-11b Tests

Test fakes, synthetic sky frames and the requirement coverage tests, which also read the Python review tool.



# Figure 4-15 CL-12 Review tool

The Python window that steps through review.xlsx: workbook access, the session with its filters and tally, the frame and metrics views, the table of rows, and the working-circle editor.



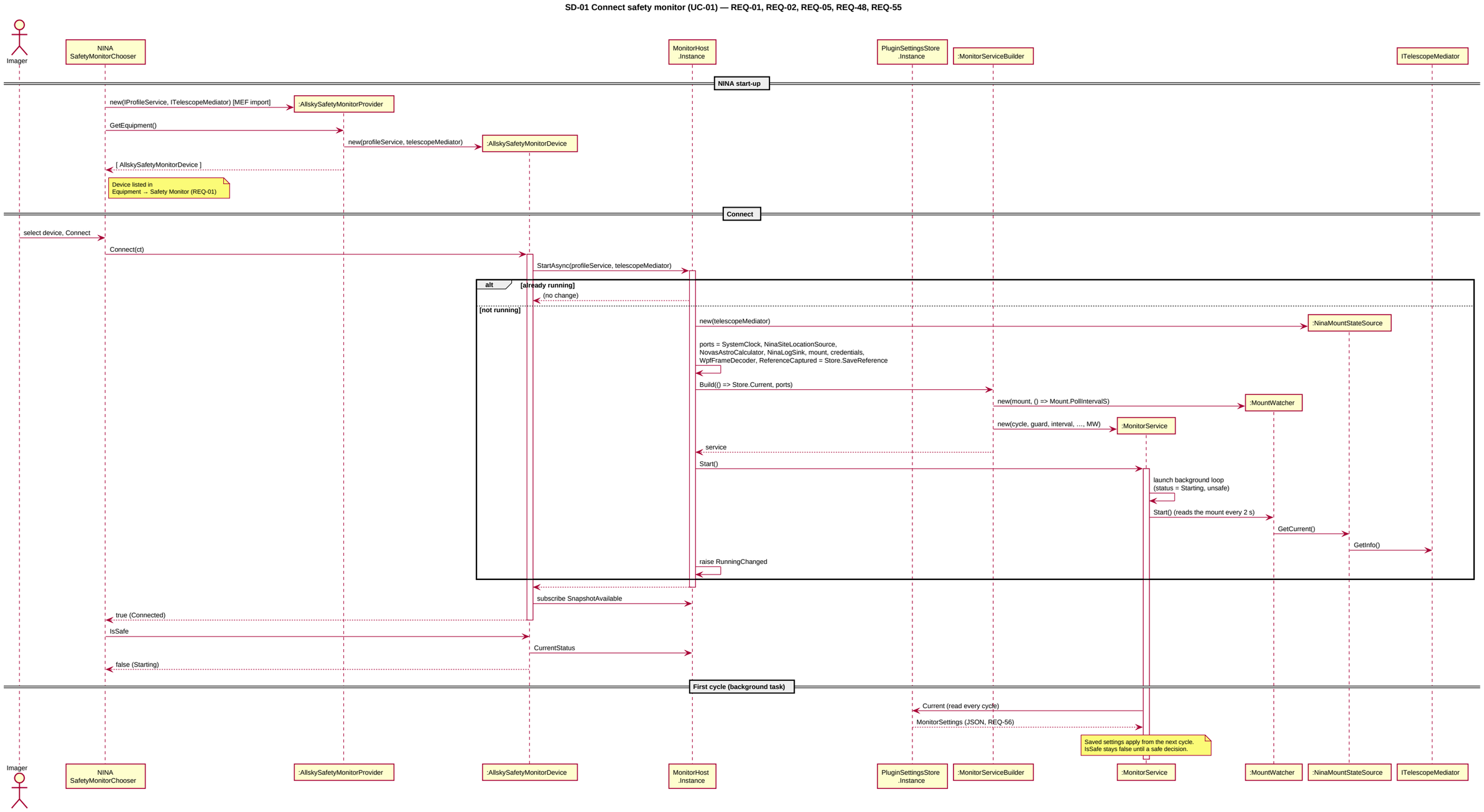
# 5 Scenarios

Fourteen sequence diagrams cover the main and failure paths. Each title names the use cases and requirements it exercises.

| Scenario | Title | Use cases |
|---|---|---|
| SD-01 | Connect safety monitor | UC-01 |
| SD-02 | Monitoring cycle, normal path | UC-04 … UC-10, UC-18 |
| SD-03 | Check the frame and detect cloud in the ROI | UC-08 |
| SD-04 | Conditions fail and clear | UC-10, UC-03 |
| SD-05 | Slewing, fallback and the working area | UC-07 |
| SD-06 | Acquire image over SFTP | UC-06 |
| SD-07 | Monitor failure and idle window | UC-19, UC-05 |
| SD-08 | Calibrate lens and define the working area | UC-12, UC-13 |
| SD-09 | Replay saved frames offline | UC-17 |
| SD-10 | Manual override and disconnect | UC-11, UC-02 |
| SD-11 | View status and edit settings | UC-14, UC-15 |
| SD-12 | View the Allsky image enlarged | UC-14 |
| SD-13 | Review the replay results | UC-20 |
| SD-14 | Adjust the working circle offline | UC-21 |

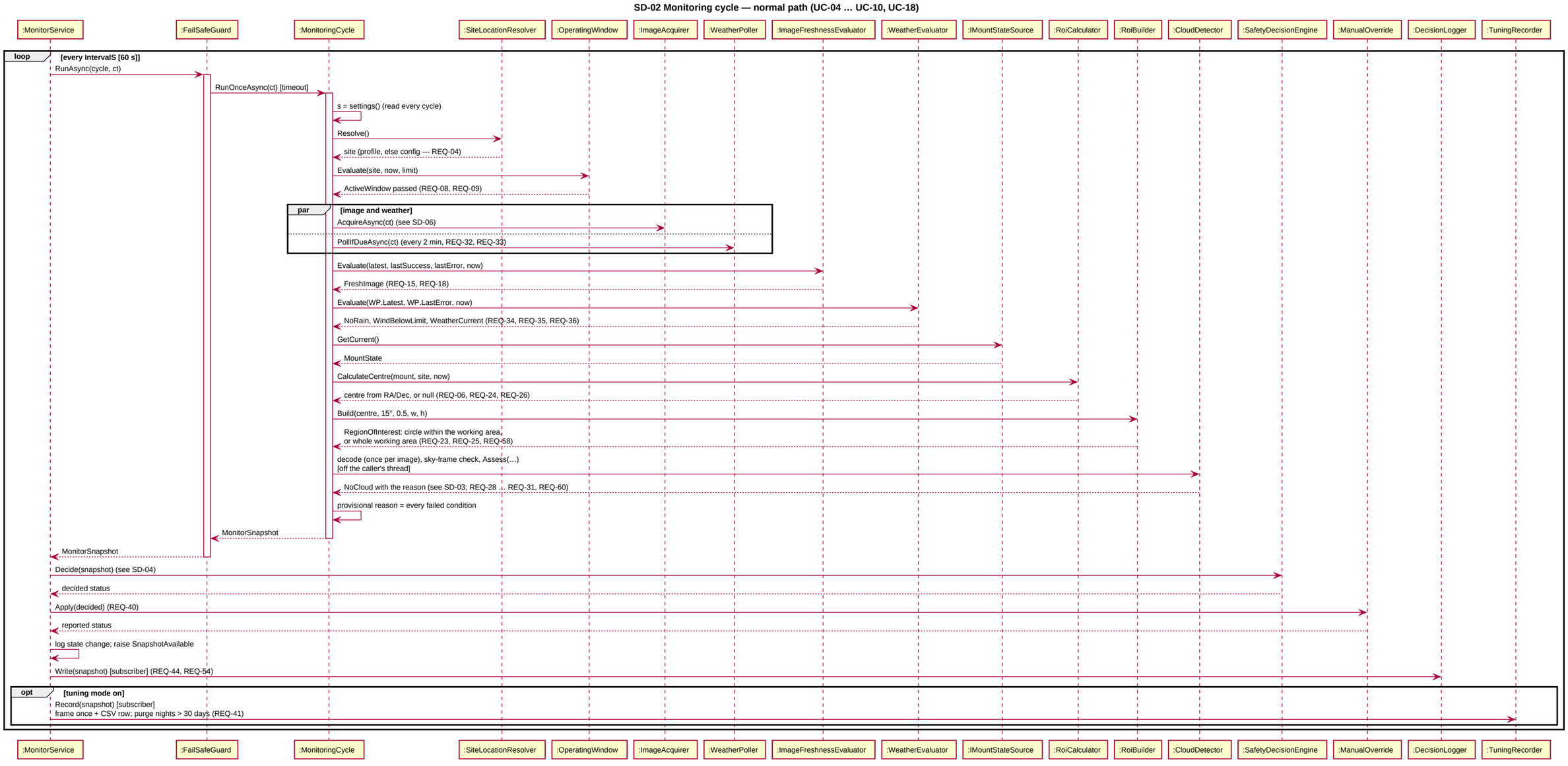
## Figure 5-1 SD-01 Connect safety monitor

NINA lists the device from the provider at start-up. Connect asks MonitorHost to start: it registers for mount updates, builds the ports and the monitor and starts its background loop. Each cycle reads the current settings. IsSafe stays false until a safe decision.



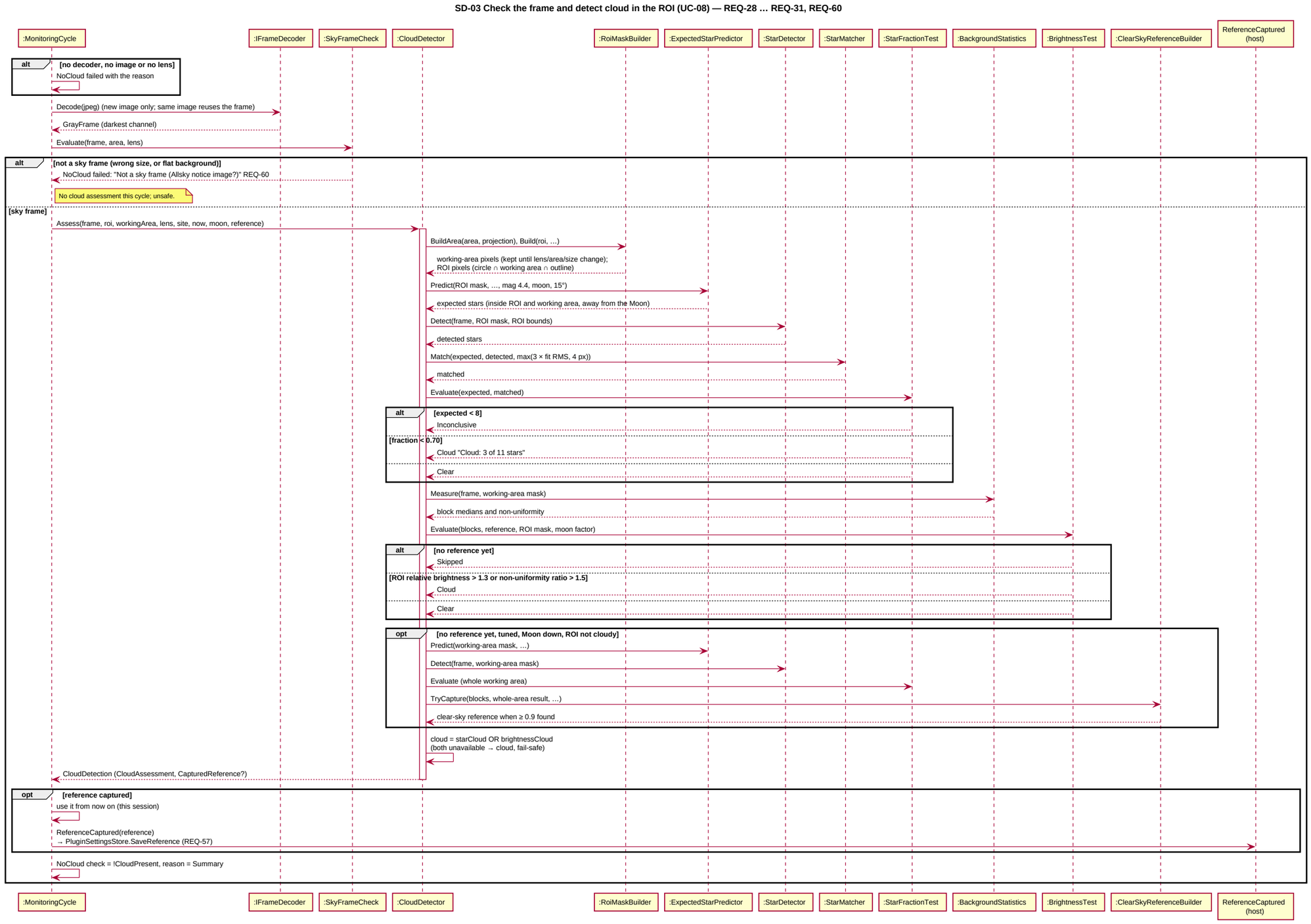
## Figure 5-2 SD-02 Monitoring cycle, normal path

One cycle: site and operating window, image and weather in parallel, freshness and weather checks, ROI and cloud; then the decision engine, override, publication and the decision log.



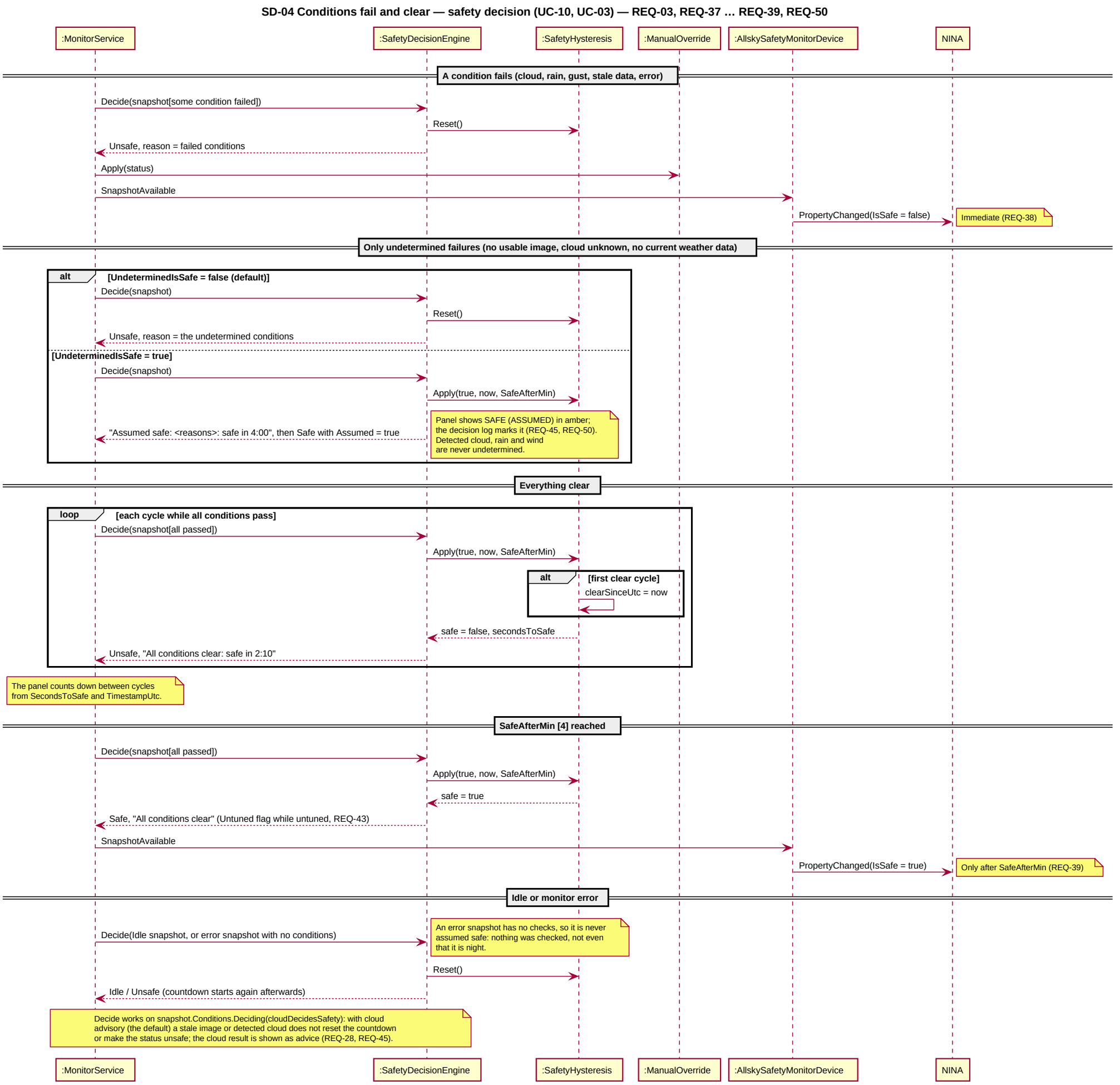
## Figure 5-3 SD-03 Check the frame and detect cloud in the ROI

Decode on the darkest channel and reject a frame that is not a sky frame; expected versus detected catalogue stars in the ROI and working area, the inconclusive case below 8 expected stars, the exposure-independent brightness test against the clear-sky map, automatic reference capture, and the fail-safe combination.



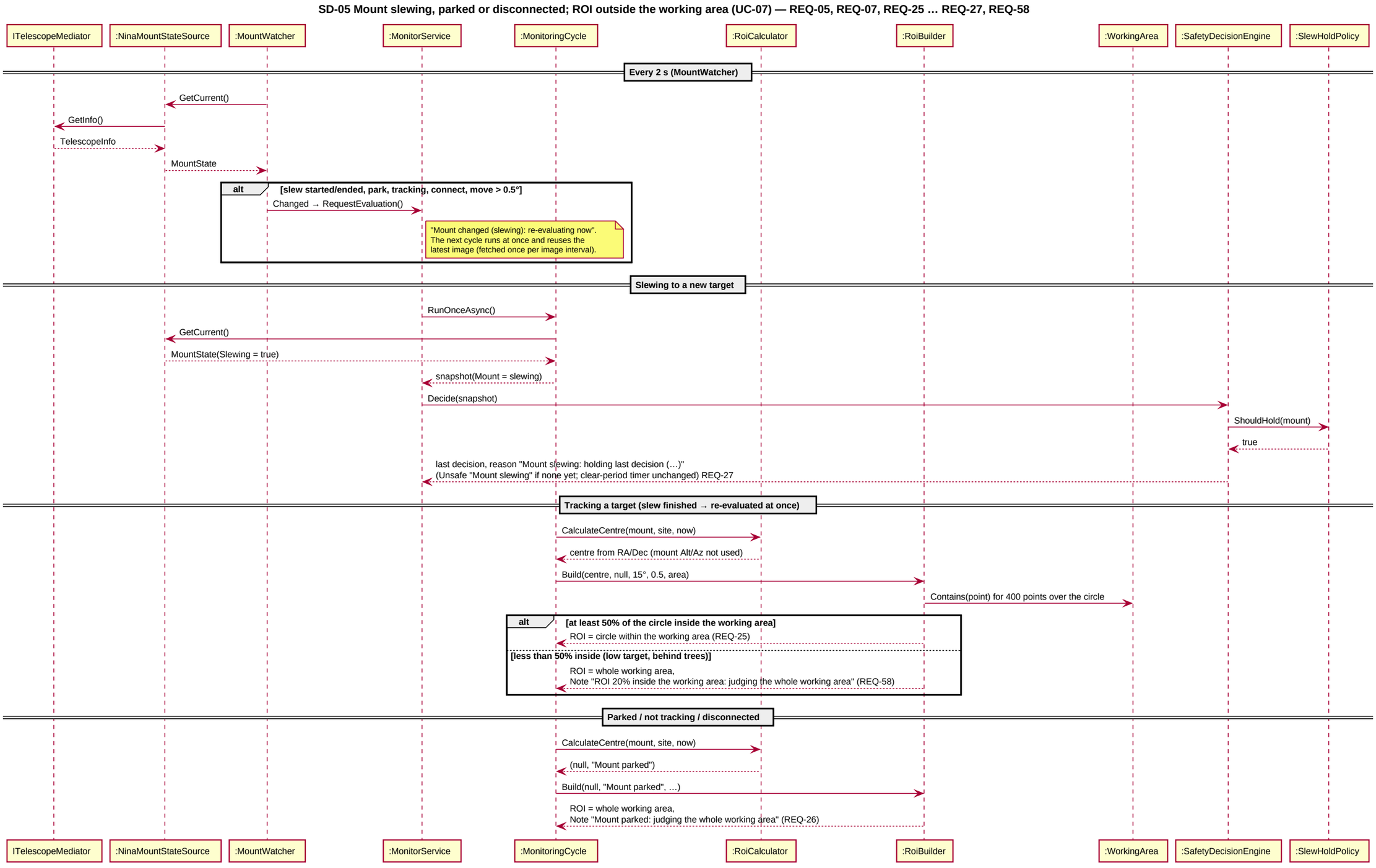
## Figure 5-4 SD-04 Conditions fail and clear

Unsafe at once; safe only after four continuous clear minutes, with a countdown. Idle or a failed cycle restarts the period.



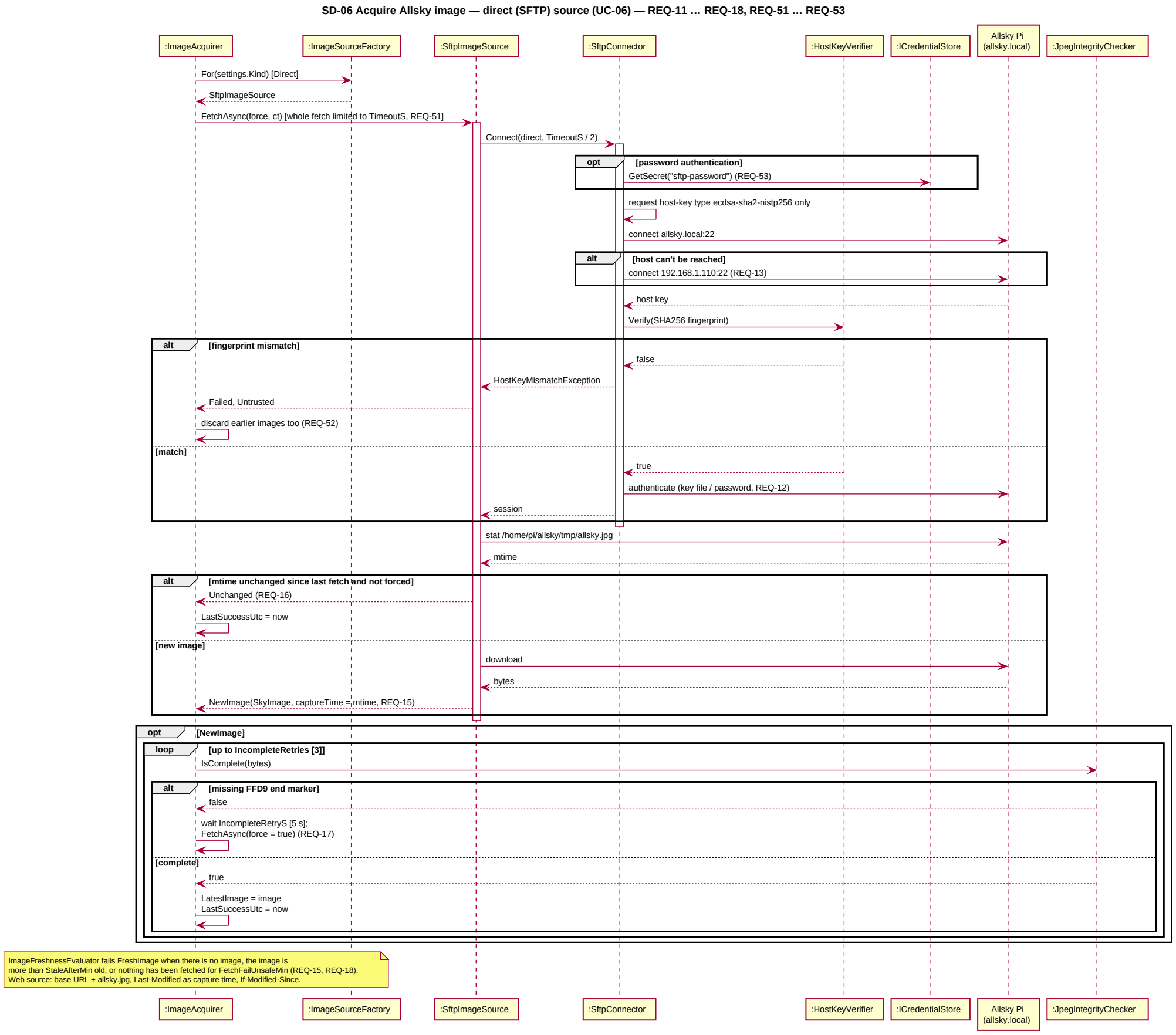
## Figure 5-5 SD-05 Slewing, fallback and the working area

MountWatcher reads the mount every 2 s and triggers an immediate re-evaluation on a change. The decision is held while slewing. The ROI is judged against the working area; with no usable mount position, or less than half the ROI inside, the whole working area is judged.



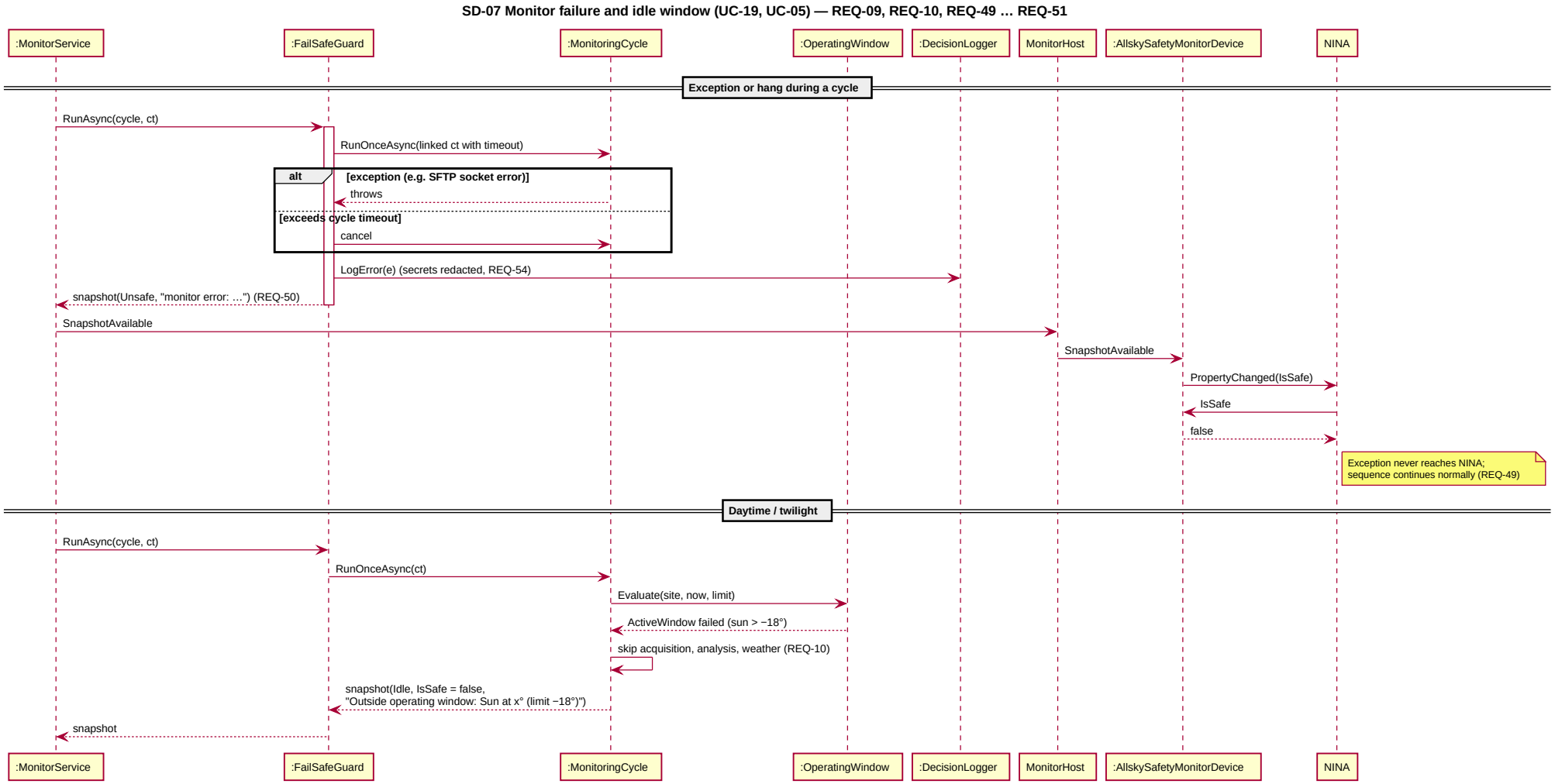
## Figure 5-6 SD-06 Acquire image over SFTP

Host-key type restriction and fingerprint check, fallback IP, stat before download, unchanged-image skip, untrusted-image discard and the incomplete-JPEG retry.



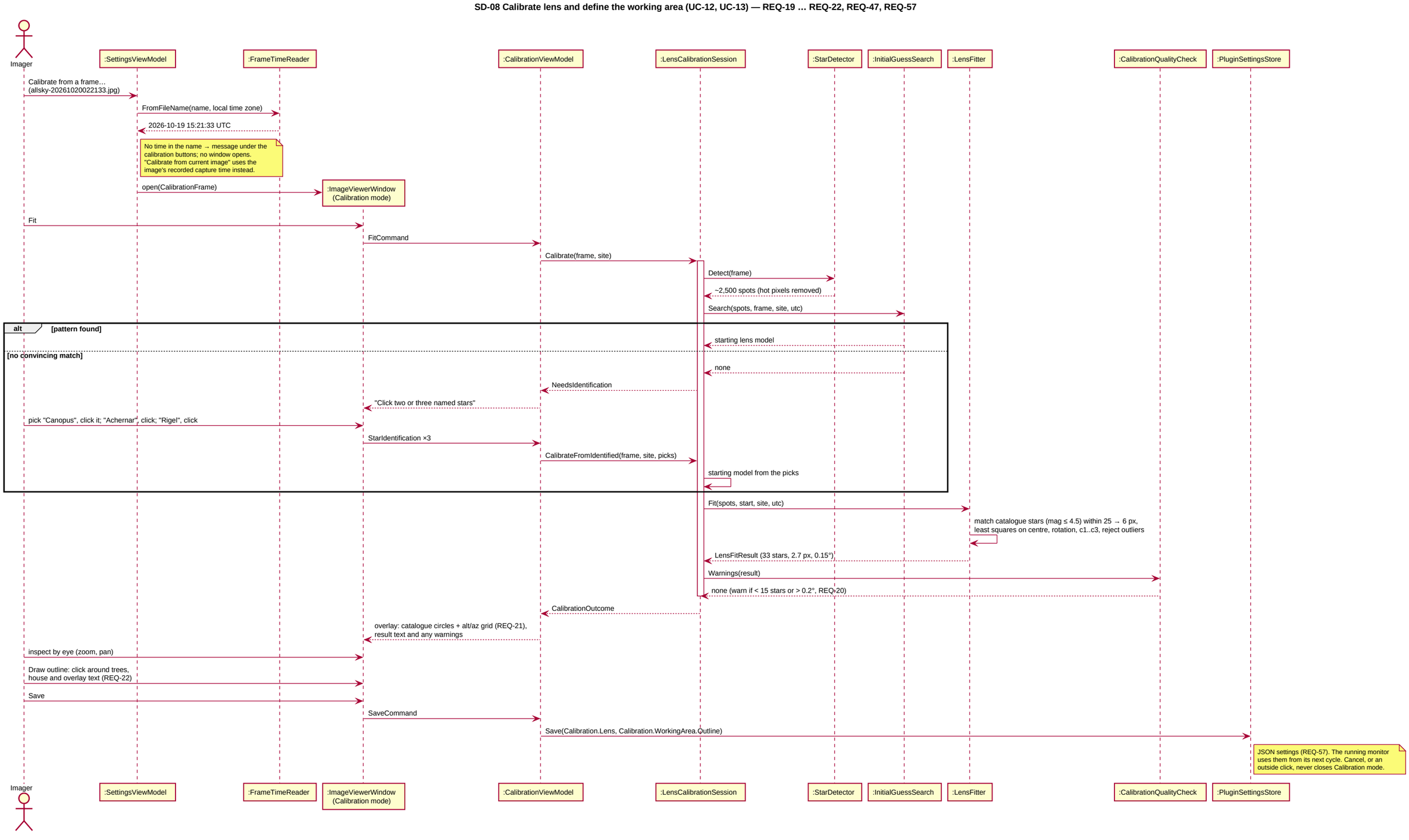
# Figure 5-7 SD-07 Monitor failure and idle window

An exception or hang becomes an Unsafe status without reaching NINA. Outside the operating window the cycle idles and reports unsafe.



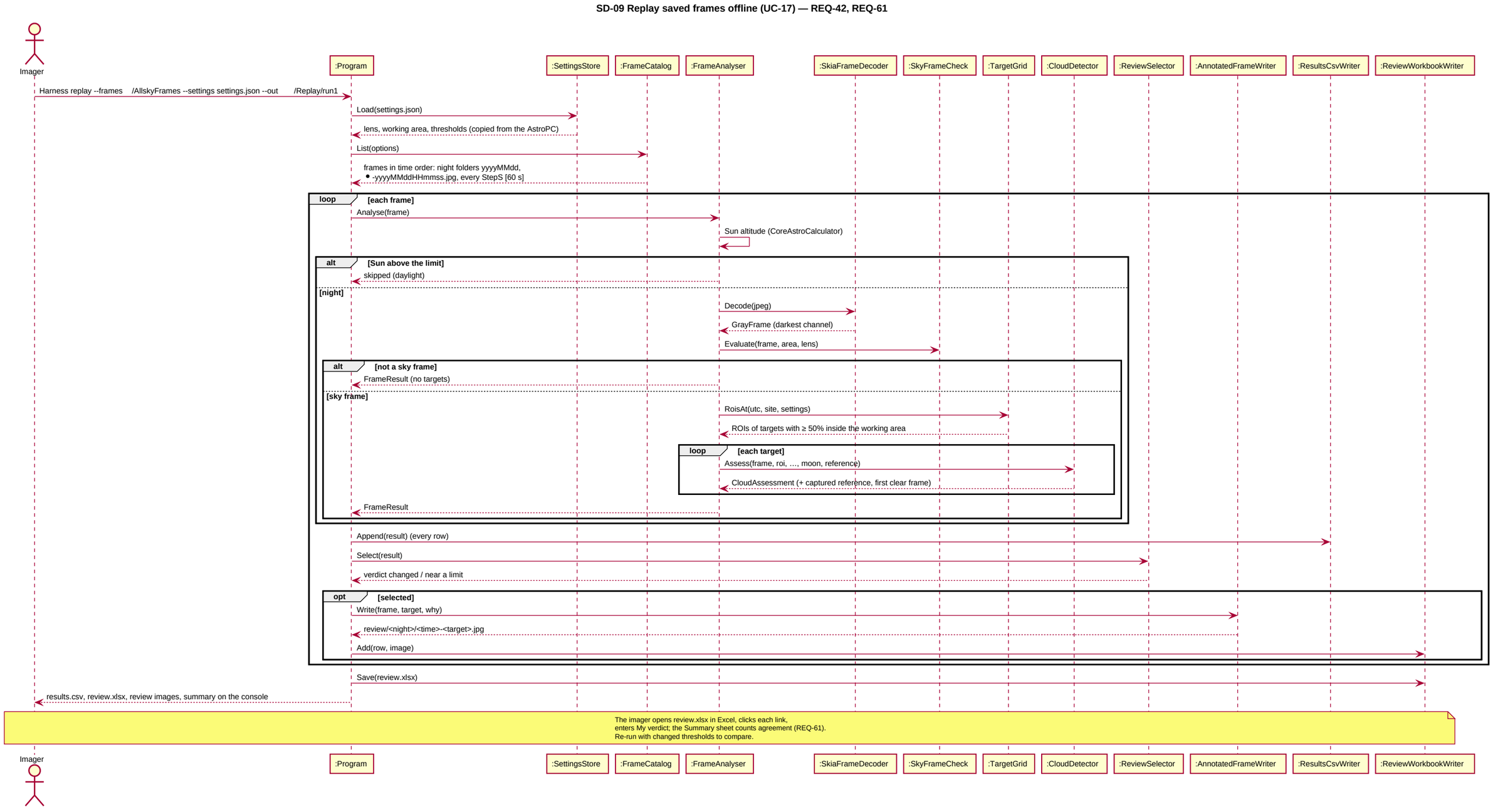
## Figure 5-8 SD-08 Calibrate lens and define the working area

From a saved clear frame (time from the file name) or the current image: detect stars, find a starting model automatically or from two or three stars the imager identifies, fit, warn on a poor fit, inspect the overlay, draw the outline, Save.



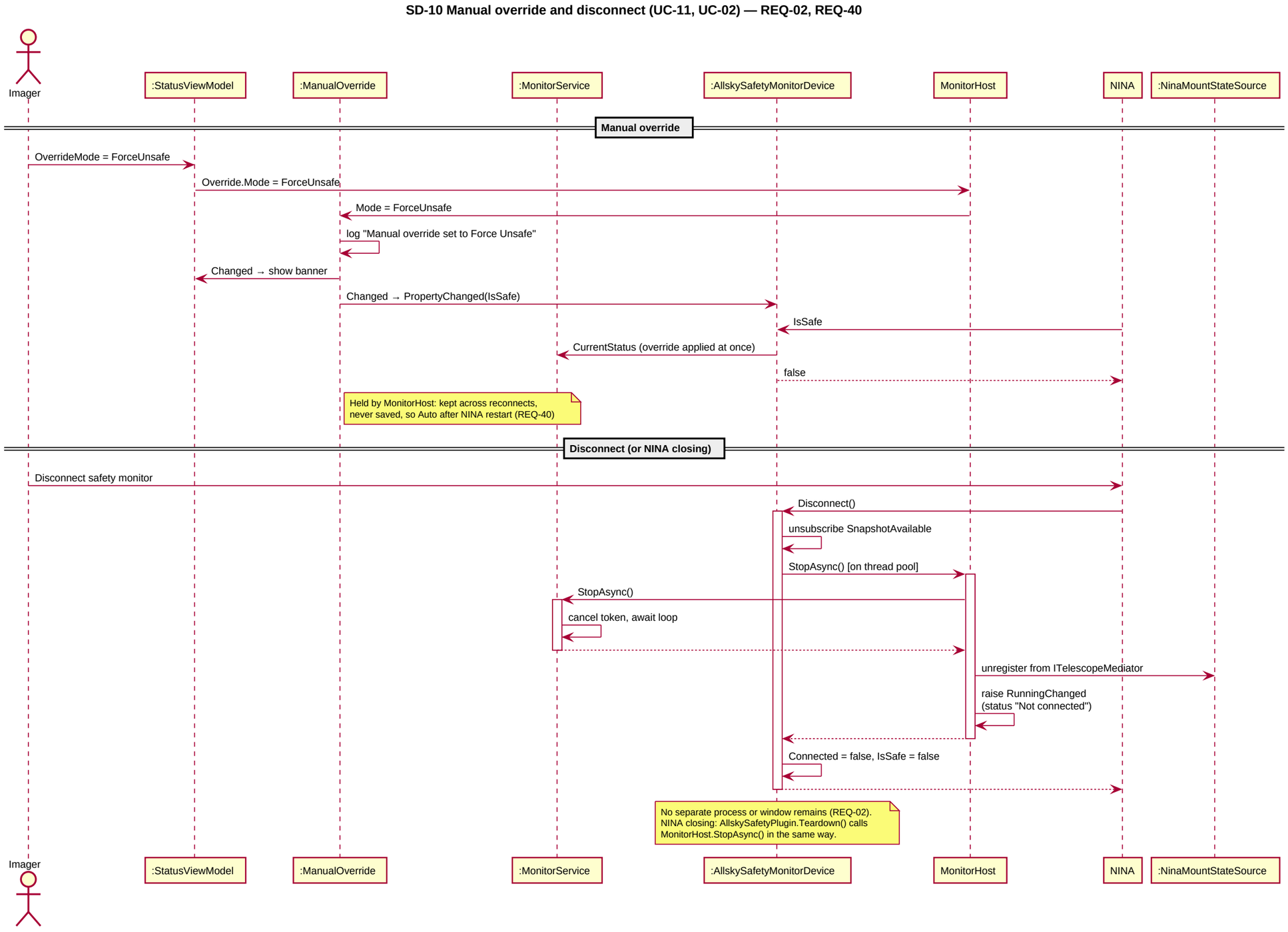
## Figure 5-9 SD-09 Replay saved frames offline

Night frames from the archive, each judged for every target inside the working area with the plugin's Core code; every row to results.csv, marked-up copies and review.xlsx where the verdict changes or a metric is near its limit.



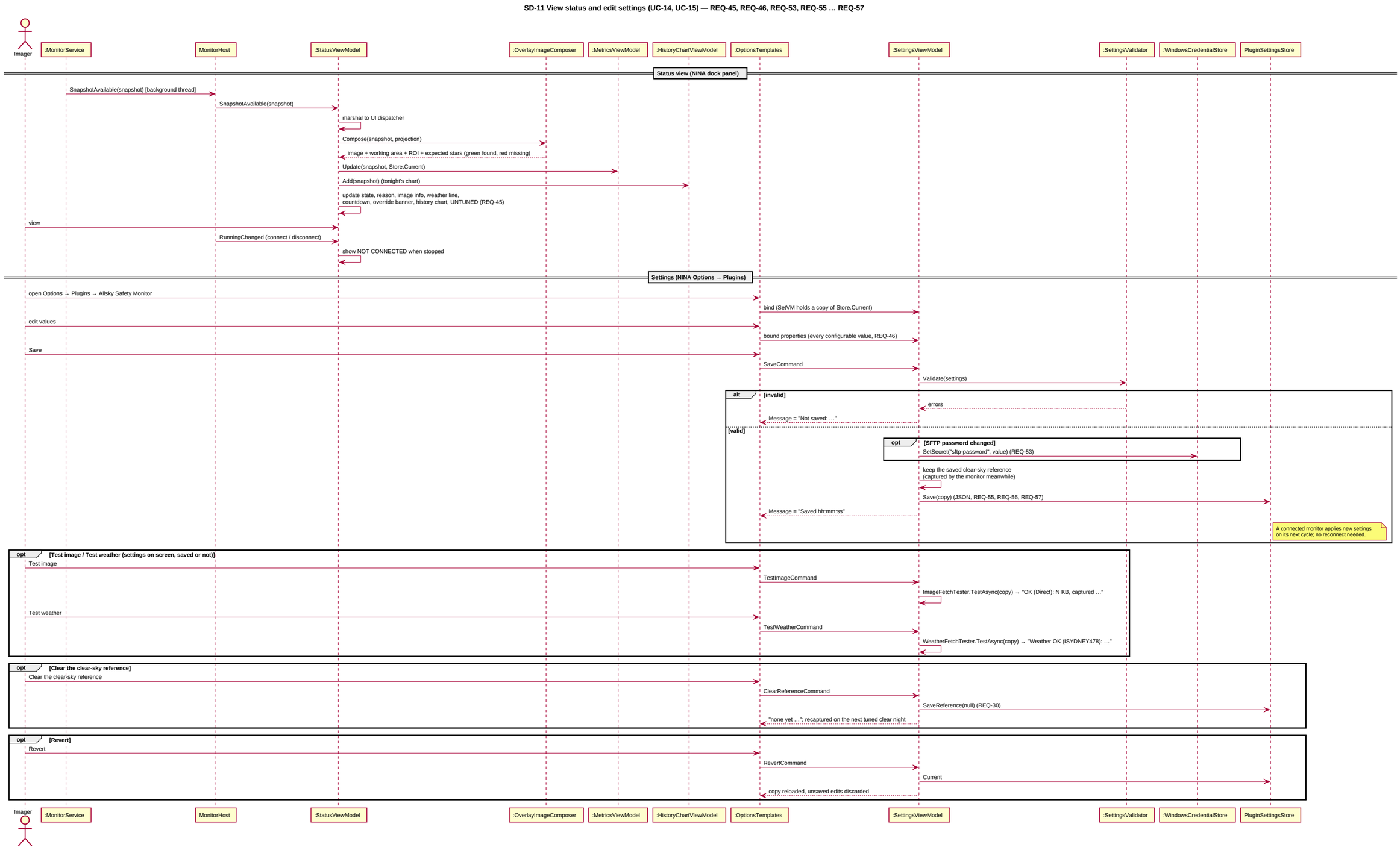
## Figure 5-10 SD-10 Manual override and disconnect

The override is held by MonitorHost, logged, applied at once and never saved. Disconnect stops the loop and unregisters from NINA, leaving nothing running.



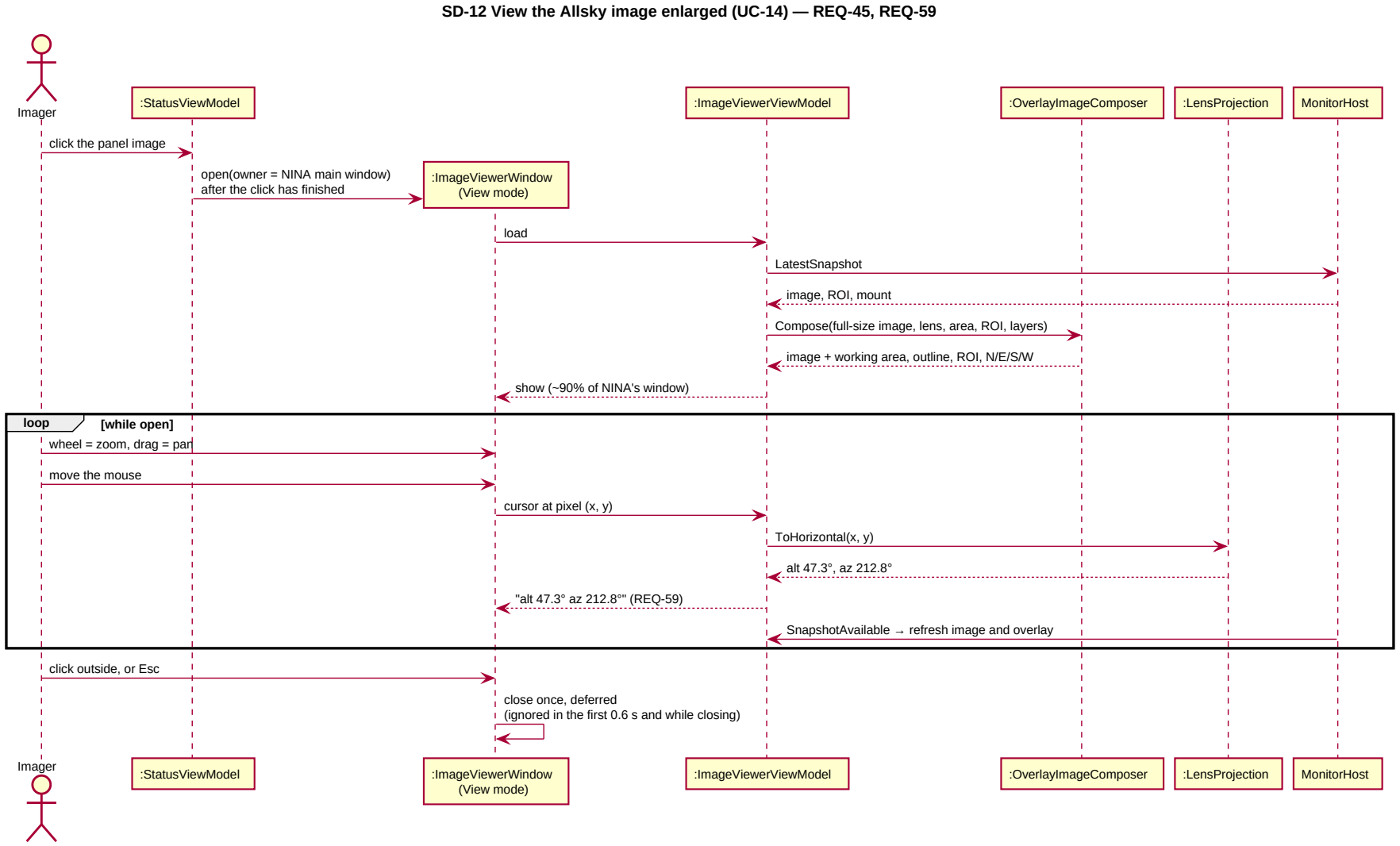
## Figure 5-11 SD-11 View status and edit settings

The status panel updates from each snapshot and every 5 s. Settings are validated and can be tested before saving; the password goes to Credential Manager and everything else to JSON.



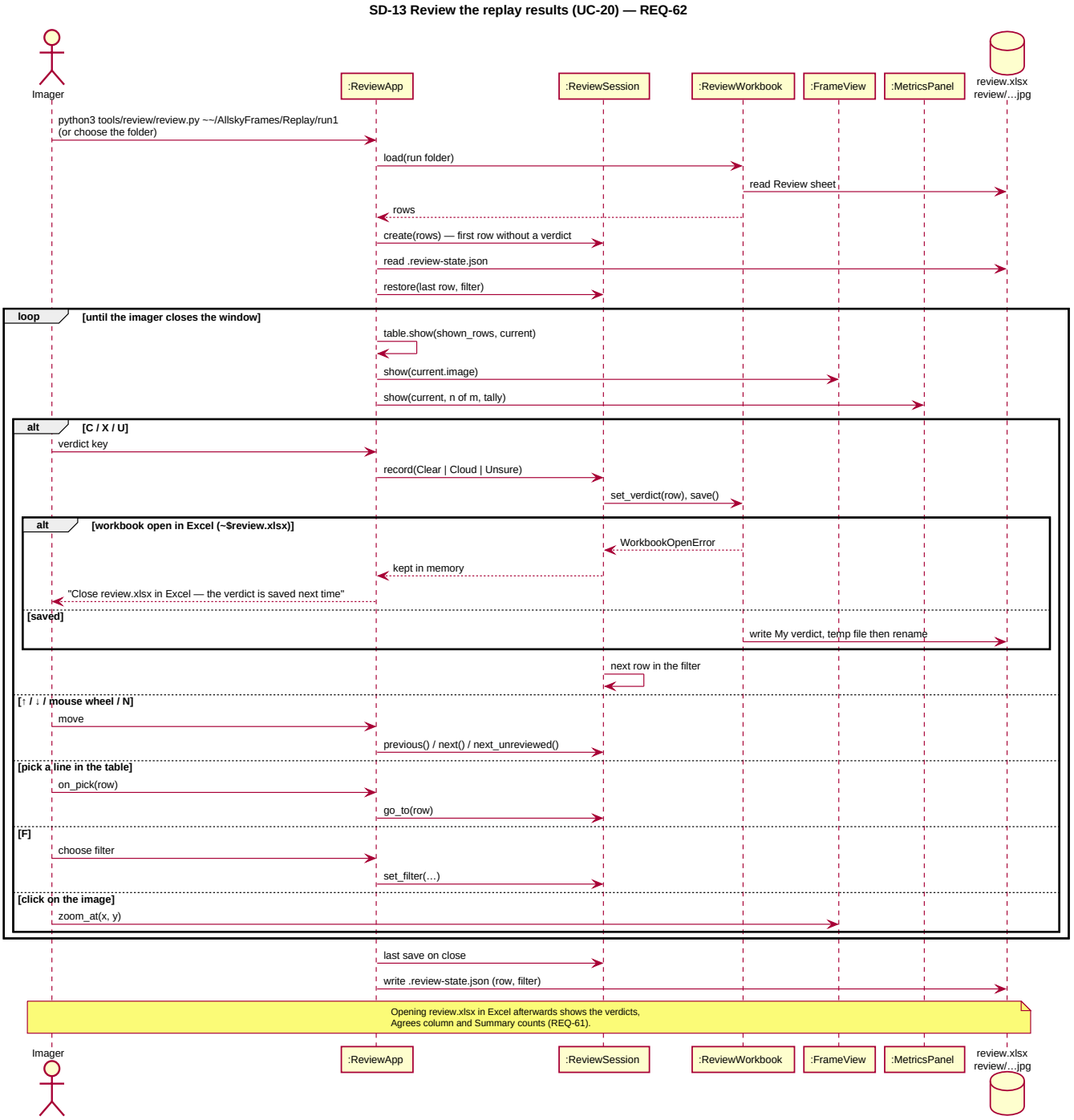
## Figure 5-12 SD-12 View the Allsky image enlarged

Clicking the panel image opens a large viewer with the overlay, zoom and pan; the cursor position is converted to altitude and azimuth with the lens model. An outside click or Esc closes it.



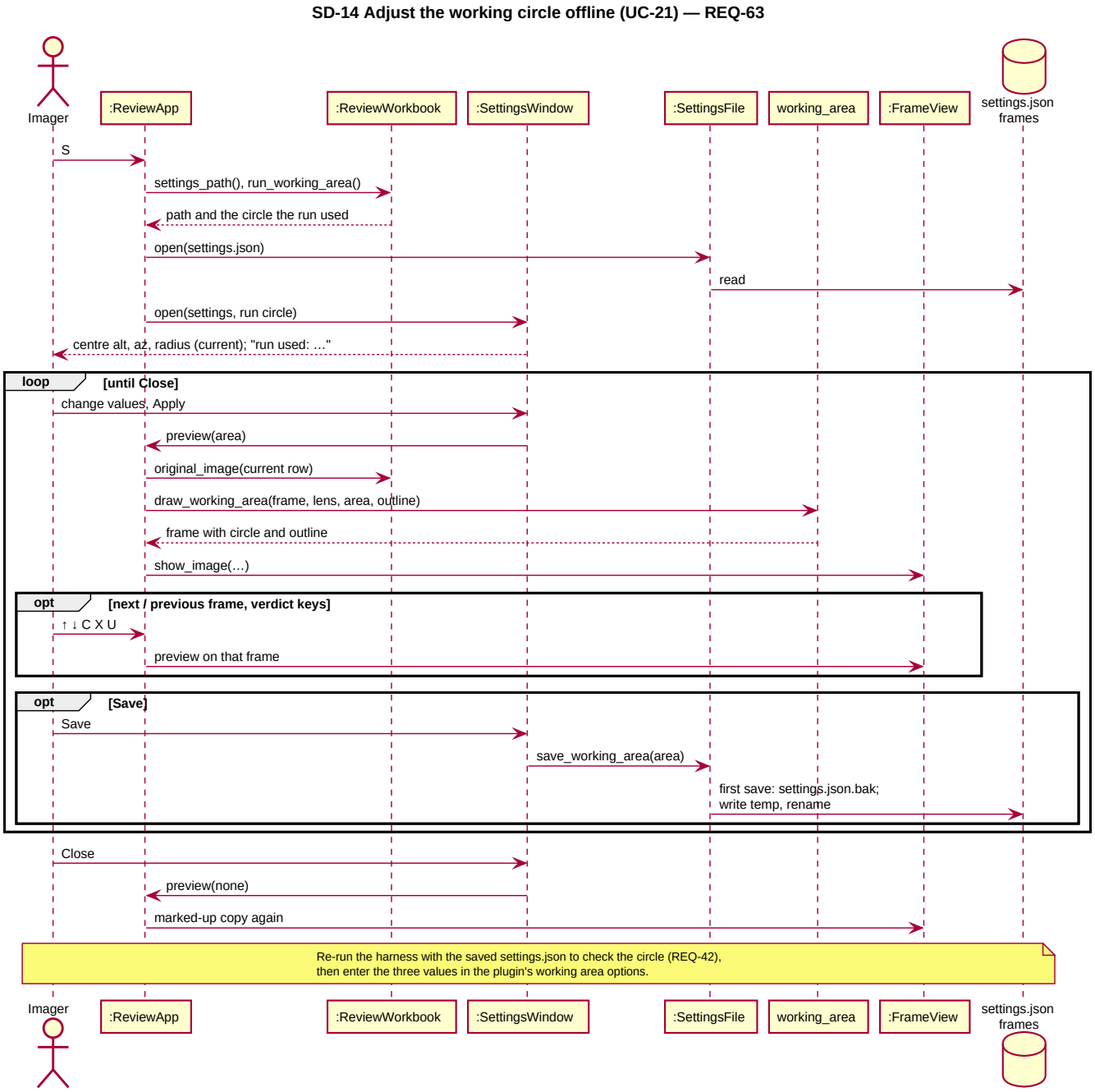
# Figure 5-13 SD-13 Review the replay results

Load review.xlsx, show each frame with its metrics, record a verdict with one key and save it at once; a save while Excel has the workbook open is refused and retried. Arrows, N and filters move through the rows; a click zooms.



## Figure 5-14 SD-14 Adjust the working circle offline

S opens the editor with the circle in settings.json and the one the run used. Apply previews the circle and outline on the original frame of any row; Save writes only the circle to settings.json. Re-run the harness to check, then enter the values in the plugin.



# 6 Traceability matrix

All 61 requirements trace to at least one use case, one design class and one scenario. The matrix is generated from the diagram sources; traceability.xlsx holds the same data with filters.

| REQ | Requirement (abridged) | Use cases | Design classes | Scenarios |
|---|---|---|---|---|
| REQ-01 | Appears in NINA as a safety monitor device (Equipment → Safety Monitor), usable by NINA sequenc… | UC-01, UC-02 | AllskySafetyMonitorDevice, AllskySafetyMonitorProvider | SD-01 |
| REQ-02 | Starts working when the device is connected in NINA and stops when it is disconnected or NINA c… | UC-01, UC-02 | AllskySafetyMonitorDevice, MonitorHost, MonitorService | SD-01, SD-10 |
| REQ-03 | Reports the result of the safety decision (§9) to NINA as safe or unsafe. | UC-03 | AllskySafetyMonitorDevice, MonitorHost, SafetyDecisionEngine, SafetyStatus | SD-04 |
| REQ-04 | The site location is taken from the active NINA profile. If that isn't available, a configured … | UC-05, UC-07 | ISiteLocationSource, NinaSiteLocationSource, SiteLocation, SiteLocationResolver | SD-02 |
| REQ-05 | The mount state is taken from NINA: connected, parked, tracking, slewing, RA (hours), Dec (degr… | UC-07 | AllskySafetyMonitorProvider, IMountStateSource, MonitorHost, MonitoringCycle, MountState, MountWatcher, NinaMountStateSource | SD-01, SD-05 |
| REQ-06 | The mount-reported Altitude/Azimuth are not used for positioning. The ASI Mount driver reports … | UC-07 | DecisionLogger, MountState, NinaMountStateSource, RoiCalculator | SD-02 |
| REQ-07 | When the mount reports no valid position (for example, disconnected), the position is treated a… | UC-07 | MountState, MountStateInterpreter, NinaMountStateSource | SD-05 |
| REQ-08 | Computes sun altitude from the site location and current time. | UC-05 | CoreAstroCalculator, IAstroCalculator, NovasAstroCalculator, OperatingWindow, SunCalculator | SD-02 |
| REQ-09 | Active only while the sun is below (astronomical twilight). | UC-05 | OperatingWindow | SD-02, SD-07 |
| REQ-10 | Outside the active window it idles: no image fetching or analysis, and it reports unsafe. | UC-05 | MonitoringCycle, OperatingWindow | SD-07 |
| REQ-11 | The image source is selectable: | UC-06 | IImageSource, ImageSourceFactory, SftpFileSession, SftpImageSource, SkyImage, WebImageSource | SD-06 |
| REQ-12 | Direct access authenticates with an SSH key (preferred,) or a password. | UC-06 | SftpConnector | SD-06 |
| REQ-13 | The Pi host is, with an optional fallback IP in case name resolution fails. | UC-06 | SftpConnector | SD-06 |
| REQ-14 | Fetches a new image every while active. | UC-06 | ImageAcquirer, MonitoringCycle | SD-06 |
| REQ-15 | Determines the image capture time (file modification time, or the web server's last-modified ti… | UC-06 | ImageFreshnessEvaluator, SftpImageSource, SkyImage, WebImageSource | SD-02, SD-06 |
| REQ-16 | For direct access, the file's modification time is checked before downloading, and the image is… | UC-06 | FetchResult, SftpImageSource | SD-06 |

| REQ | Requirement (abridged) | Use cases | Design classes | Scenarios |
|---|---|---|---|---|
| REQ-17 | An image caught part-way through being written (an incomplete JPEG) is discarded and fetched ag… | UC-06 | ImageAcquirer, JpegIntegrityChecker | SD-06 |
| REQ-18 | If no image can be fetched for, reports unsafe. | UC-06 | ImageAcquirer, ImageFreshnessEvaluator | SD-02, SD-06 |
| REQ-60 | An image that is not a sky frame, such as the notice image Allsky shows when its software is no… | UC-06 | LensCalibration, MonitorSnapshot, MonitoringCycle, SkyFrameCheck | SD-02, SD-03 |
| REQ-19 | Provides a one-off calibration that maps alt/az to Allsky image pixel positions: image centre, … | UC-12 | CalibrationViewModel, LensCalibration, LensCalibrationSession, LensFitter, LensProjection | SD-08 |
| REQ-20 | Calibration is done by matching identified catalogue stars in a real clear-sky frame. A calibra… | UC-12 | CalibrationQualityCheck, CalibrationViewModel, FrameTimeReader, IFrameDecoder, InitialGuessSearch, LensCalibrationSession, LensFitSettings, LensFitter, SkyStars, StarCatalogue, StarDetector, WpfFrameDecoder | SD-08 |
| REQ-21 | Shows a verification overlay (catalogue stars and an alt/az grid drawn on the live image) so ca… | UC-12 | CalibrationViewModel, ImageViewerViewModel, OverlayImageComposer | SD-08 |
| REQ-22 | A working area defines the part of the sky the monitor can see and assess. It is a circle of co… | UC-13 | CalibrationViewModel, OverlayImageComposer, SettingsValidator, SettingsViewModel, WorkingArea, WorkingAreaDefinition | SD-08 |
| REQ-23 | The ROI is a circle of radius on the sky, centred on the mount's current RA/Dec. | UC-07 | OverlayImageComposer, RegionOfInterest, RoiBuilder, RoiCalculator, RoiSettings, SettingsViewModel | SD-02 |
| REQ-24 | RA/Dec is converted to alt/az using the coordinate epoch (JNOW on this rig), the site location … | UC-07 | CoordinateConverter, CoreAstroCalculator, IAstroCalculator, LensProjection, NovasAstroCalculator, RoiCalculator, SkyStars | SD-02 |
| REQ-25 | Any part of the ROI outside the working area (REQ-22) is excluded from analysis. | UC-07 | RegionOfInterest, RoiBuilder, RoiMaskBuilder, WorkingArea | SD-02, SD-05 |
| REQ-58 | If less than a configurable fraction of the ROI lies inside the working area, the whole working… | UC-07 | RegionOfInterest, RoiBuilder, RoiSettings, SettingsValidator, SettingsViewModel, StatusViewModel | SD-02, SD-05 |
| REQ-26 | The ROI falls back to the whole working area when the mount is not connected, is parked, is not… | UC-07 | MountStateInterpreter, RegionOfInterest, RoiBuilder, RoiCalculator | SD-02, SD-05 |
| REQ-27 | While the mount is slewing, the current safety state is held and the ROI is updated once the sl… | UC-07 | MonitorService, MountSettings, MountWatcher, SafetyDecisionEngine, SettingsValidator, SettingsViewModel, SlewHoldPolicy | SD-05 |
| REQ-28 | Cloud detected in the ROI is always shown to the imager. It makes the status unsafe only when t… | UC-08 | CloudAssessment, CloudDetection, CloudDetector, DecisionSettings, MonitoringCycle, SafetyConditions, SafetyDecisionEngine | SD-02, SD-03, SD-04 |
| REQ-29 | Primary test: predict which catalogue stars should be visible in the ROI (down to a limiting ma… | UC-08 | CloudDetectionSettings, ExpectedStarPredictor, RoiMaskBuilder, SettingsMigrator, StarCatalogue, StarDetector, StarFractionTest, StarMatcher | SD-03 |
| REQ-30 | Secondary test, comparing the ROI with a clear-sky reference: a map of sky brightness and non-u… | UC-08 | BackgroundStatistics, BrightnessTest, ClearSkyReference, ClearSkyReferenceBuilder, CloudDetection, CloudDetectionSettings, PluginSettingsStore | SD-03, SD-11 |

| REQ | Requirement (abridged) | Use cases | Design classes | Scenarios |
|---|---|---|---|---|
| REQ-31 | Stars within of the Moon are excluded from the expected list, and brightness thresholds are adj… | UC-08 | BrightnessTest, CloudDetectionSettings, CoreAstroCalculator, ExpectedStarPredictor, IAstroCalculator, MoonBrightnessAdjuster, MoonCalculator, MoonState, NovasAstroCalculator | SD-02, SD-03 |
| REQ-32 | Polls the Weather.com PWS current-observations API for station every, with metric units, authen… | UC-09 | IWeatherProvider, PwsWeatherClient, WeatherPoller | SD-02 |
| REQ-33 | Values are read from observations: epoch (observation time), metric.precipRate (mm/h), metric.w… | UC-09 | PwsObservationParser, WeatherObservation | SD-02 |
| REQ-34 | Any precipitation rate above zero → unsafe immediately. | UC-09 | WeatherEvaluator | SD-02 |
| REQ-35 | Wind gust above → unsafe. | UC-09 | WeatherEvaluator | SD-02 |
| REQ-36 | If the station data is older than or unavailable → unsafe. Weather checking can be switched off… | UC-09 | WeatherEvaluator | SD-02 |
| REQ-37 | The overall status is safe only when all of these are true: active window, no rain, wind below … | UC-10 | ConditionCheck, MonitoringCycle, SafetyConditions, SafetyDecisionEngine | SD-04 |
| REQ-38 | Transition to unsafe is immediate when any condition fails. | UC-10 | MonitorService, SafetyDecisionEngine, SafetyHysteresis | SD-04 |
| REQ-39 | Transition to safe happens only after all conditions have held continuously for. | UC-10 | DecisionSettings, SafetyDecisionEngine, SafetyHysteresis, SafetyStatus | SD-04 |
| REQ-40 | A manual override (Force Safe / Force Unsafe / Auto) is provided, shown prominently and recorde… | UC-11 | ManualOverride, MonitorService, SafetyStatus, StatusViewModel | SD-02, SD-10 |
| REQ-41 | A tuning mode saves every frame together with its computed metrics, in a format readable by Exc… | UC-16 | MetricsCsvWriter, MonitorSnapshot, TuningRecorder, TuningSettings | SD-02 |
| REQ-42 | Saved frames can be re-analysed offline, without the rig or a live NINA session, to test thresh… | UC-17 | CoreAstroCalculator, FrameAnalyser, FrameCatalog, Program, ProgressLine, ReplayOptions, ResultsCsvWriter, SkiaFrameDecoder, TargetGrid | SD-09, SD-14 |
| REQ-61 | The results of an offline re-analysis can be checked by eye: a row per frame and sky position w… | UC-17 | AnnotatedFrameWriter, ReplayOptions, ResultsCsvWriter, ReviewSelector, ReviewWorkbook, ReviewWorkbookWriter | SD-09, SD-13 |
| REQ-62 | The frames picked for review (REQ-61) can be gone through one at a time: each marked-up frame i… | UC-20 | FrameView, MetricsPanel, ReviewApp, ReviewRow, ReviewSession, ReviewTable, ReviewWorkbook, RowFilter, Tally | SD-13 |
| REQ-63 | The working area (REQ-22) can be adjusted away from the rig: its centre and radius can be chang… | UC-21 | LensModel, ReviewApp, ReviewWorkbook, SettingsFile, SettingsWindow, WorkingArea, WorkingCircleDrawing | SD-14 |
| REQ-43 | Until the thresholds have been tuned, conservative defaults (biased towards unsafe) are used an… | UC-16 | CloudDetectionSettings, SafetyStatus, StatusViewModel, TuningStateEvaluator | SD-04 |
| REQ-44 | Keeps a rolling log of state changes, metrics, errors and the reason for every safe/unsafe deci… | UC-18 | DecisionLogger | SD-02 |

| REQ | Requirement (abridged) | Use cases | Design classes | Scenarios |
|---|---|---|---|---|
| REQ-45 | A status view shows: | UC-14 | CloudAssessment, HistoryChartViewModel, MetricsViewModel, MonitorSnapshot, OverlayImageComposer, SafetyStatus, StatusViewModel | SD-04, SD-11, SD-12 |
| REQ-46 | A settings view edits every configurable value. | UC-15 | AllskySafetyPlugin, ImageFetchTester, SettingsValidator, SettingsViewModel, WeatherFetchTester | SD-11 |
| REQ-47 | A calibration view supports lens calibration (REQ-19–21) and defining the working area (REQ-22)… | UC-12, UC-13 | CalibrationViewModel | SD-08 |
| REQ-59 | The Allsky image, with its overlay, can be viewed enlarged, and the altitude and azimuth of any… | UC-14 | ImageViewerViewModel, StatusViewModel | SD-12 |
| REQ-48 | Runs on the Windows AstroPC within NINA 3.x (currently the 3.3 nightly). | UC-01 | AllskySafetyPlugin | SD-01 |
| REQ-49 | A failure in the safety monitor (crash, hang, network fault) must not disrupt NINA or a running… | UC-19 | AllskySafetyMonitorDevice, FailSafeGuard, MonitorService | SD-07 |
| REQ-50 | Fail-safe: any failure, or any state that can't be determined, results in unsafe, or in safe wh… | UC-19 | ConditionCheck, DecisionSettings, FailSafeGuard, ImageFreshnessEvaluator, SafetyConditions, SafetyDecisionEngine, SafetyStatus, WeatherEvaluator | SD-04, SD-07 |
| REQ-51 | Every network operation (image fetch, weather poll) has a timeout, so a stalled connection can'… | UC-19 | FailSafeGuard, PwsWeatherClient, SftpConnector, SftpImageSource, WebImageSource | SD-06, SD-07 |
| REQ-52 | The Pi's host key is verified against a pinned fingerprint; a mismatch is reported and the imag… | UC-06 | FetchResult, HostKeyMismatchException, HostKeyVerifier, SftpConnector | SD-06 |
| REQ-53 | An SFTP password, if used, is stored using Windows' credential storage, never in the settings f… | UC-06, UC-15 | CredentialKeys, ICredentialStore, SettingsViewModel, SftpConnector, WindowsCredentialStore | SD-06, SD-11 |
| REQ-54 | Passwords and API keys are never written to the log. | UC-18 | DecisionLogger, PwsWeatherClient, RedactingLogSink, SecretRedactor | SD-02, SD-07 |
| REQ-55 | All configurable values persist between sessions. | UC-15 | MonitorSettings, PluginSettingsStore, SettingsMigrator, SettingsStore, SettingsViewModel | SD-01, SD-11 |
| REQ-56 | Settings are stored in a JSON file. | UC-15 | PluginSettingsStore, SettingsStore | SD-01, SD-11 |
| REQ-57 | Calibration results (REQ-19), the working area (REQ-22) and the clear-sky reference (REQ-30) ar… | UC-15 | CalibrationData, CalibrationViewModel, ClearSkyReference, PluginSettingsStore, WorkingAreaDefinition | SD-03, SD-08, SD-11 |